MANUFACTURING & SERVICE OPERATIONS MANAGEMENT
https://pubsonline.informs.org/journal/msom
Vol. 00, No. 0, Xxxxx 0000, pp. 000–000
ISSN 1523-4614, EISSN 1526-5498

Submitted to *Manufacturing & Service Operations Management*

How to (and How Not to) Increase Free-Shipping Thresholds: Evidence from Online Retail

Quan Zhou
Desautels Faculty of Management, McGill University, quan.zhou3@mail.mcgill.ca

Qianran (Jenny) Jin
CUHK Business School, The Chinese University of Hong Kong, jennyjin@cuhk.edu.hk

Mehmet Gümüş
Desautels Faculty of Management, McGill University, mehmet.gumus@mcgill.ca

Animesh Animesh
Desautels Faculty of Management, McGill University, animesh.animesh@mcgill.ca

Abstract. **Problem definition:** Contingent free shipping (CFS) policies are widely used by e-commerce platforms to encourage larger cart values, yet platforms increasingly struggle with how to adjust these thresholds in response to rising fulfillment costs. This paper examines how increasing the CFS threshold affects consumer behavior and derives conditions under which such an increase is optimal for the platform. **Methodology:** Using detailed transaction and clickstream data from a North American online grocery platform, we analyze a threshold increase from \$80 to \$100 by employing a difference-in-differences design. **Results:** We find that raising the CFS threshold results in a 19.4% decline in consumer spending, driven primarily by a reduction in average cart value and, to a lesser extent, a decline in purchase frequency. Consumers also purchase fewer distinct products, shift to lower-priced items, and reduce their share of spending on popular products. We identify two mechanisms underlying these responses. First, the original threshold serves as a reference point, making the new threshold feel less attainable and weakening top-up behavior. Second, consumers with high search effort face disproportionate difficulty reaching the higher threshold and are more likely to reduce or abandon their basket. **Managerial implications:** To assess the implications for optimal policy design, we develop a stylized analytical model that captures both cost- and demand-side factors associated with increasing the CFS threshold. We find that it is optimal to increase the CFS threshold only when the average cart value is sufficiently close to the original threshold. This result aligns closely with our empirical findings and remains robust to extensions that incorporate consumer heterogeneity.

Key words: e-commerce, contingent free shipping, consumer purchase behavior, difference-in-differences

1. Introduction

E-commerce platforms have experienced explosive growth over the past decade. As of 2025, 2.77 billion consumers worldwide shopped online, generating more than $6.3 trillion in sales, accounting for 20.1% of global retail purchases (Seller Commerce 2025a). This share is projected to reach 25% by 2030 (Statista 2025). Despite these impressive growth figures, online retailers continue to face a persistent challenge: converting website visitors into paying consumers. Many potential buyers abandon their carts before completing a purchase. A leading reason for checkout abandonment is the presence of additional costs, most notably shipping surcharges (Baymard Institute 2025).

To address this challenge, online retailers use different shipping policies to balance delivery costs and demand. While some charge a shipping fee (either a flat fee regardless of order size or distance-based/weight-based fees that vary with logistics costs), others offer unconditional free shipping for all consumers or only for paid members. A third shipping policy that is becoming popular is contingent free shipping (CFS), where the retailer waives the shipping fee only if a consumer's order exceeds a predefined spending threshold.

The rationale behind CFS policies is twofold. From a cost perspective, aggregating larger orders allows online platforms to reduce per-unit delivery costs through economies of scale. From a demand perspective, by offering free delivery contingent on meeting a spending threshold, CFS policies incentivize consumers to add items to their carts or switch to higher-priced products, a phenomenon commonly referred to as top-up behavior. The effectiveness of CFS in shaping consumer purchase behavior has been documented in industry practice (Brandt 2018) and examined in academic research (Cachon et al. 2018, Chen and Ngwe 2018, Hemmati et al. 2021, Kabra et al. 2024, Lewis et al. 2006). Given these benefits, it is no surprise that approximately 65% of online retailers today employ some form of contingent free shipping to reduce shipping frictions and attract more consumers (Seller Commerce 2025b).

However, to manage rising fulfillment costs, many of these platforms periodically raise the free-shipping threshold (Sancia 2025), with the average CFS threshold rising from approximately $52 in 2019 to $103 in 2025 (Nigrelli 2025). From a cost perspective, such adjustments are well justified: increasing the threshold reduces the share of orders in which the platform bears shipping costs. Yet as thresholds continue to rise, nearly doubling in recent years, a critical question remains: while CFS policies effectively encourage purchases at a given threshold, does this effectiveness persist when the threshold is raised? Or do higher thresholds fundamentally alter consumer behavior, undermining the effectiveness of CFS altogether?

Both academic research and managerial practice lack clear answers. For instance, Amazon Fresh increased its free-shipping threshold from \$35 to \$150 in February 2023, only to reduce it to \$100 ten months later (Moran 2023). Similarly, Freshippo, an online grocery platform operated by Alibaba Group, raised its threshold from 49 RMB to 99 RMB in February 2024, but reverted to the original level within two months (Chen 2024). These rapid reversals suggest that the adjustments did not perform as expected, underscoring how little is understood about whether and when threshold increases remain effective.

Motivated by this gap, the central objective of this paper is to study the demand-side effects of increasing CFS thresholds. We address three interrelated research questions:

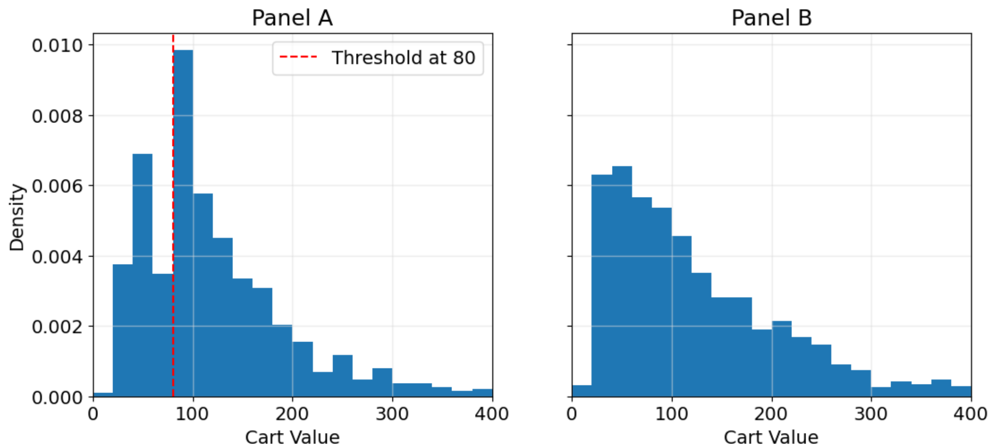
1. *What is the overall impact of raising the CFS threshold on consumer purchase behavior?*
2. *What mechanisms explain consumers' responses to threshold increases?*
3. *What are the implications for optimal threshold design?*

To answer these questions, we analyze a unique panel dataset from our industrial partner, which is a North American e-commerce platform that connects more than 1,000 farmers and food processors with end consumers. Like many online grocery retailers, this platform faced high delivery costs. To exploit economies of scale in fulfillment, the platform introduced a CFS policy for consumers living in metropolitan areas, offering them free delivery for orders that exceed \$80. In contrast, consumers residing elsewhere are subject to standard paid shipping.

Using cart-level transaction data from the collaborating platform, we confirm that the top-up behavior also arises in our empirical setting. Figure 1 presents the distribution of completed cart values under these two regimes. Panel A shows a pronounced mass of orders just above the \$80 free-shipping threshold in CFS regions, accompanied by relatively fewer orders immediately below \$80, which is clear evidence of top-up behavior. In contrast, Panel B shows no such concentration when free shipping is unavailable. These patterns align with prior studies, indicating that the existing CFS threshold effectively encourages consumers to increase their cart values.

Nevertheless, the platform increased the CFS threshold from \$80 to \$100 in November 2023 for consumers in metropolitan regions, while leaving shipping policies unchanged elsewhere. This policy change creates a natural difference-in-differences (DiD) setting spanning nearly two years. Namely, between January 2023 and October 2023, consumers in metropolitan regions faced an \$80 threshold; from November 2023 onward, the threshold increased to \$100. In addition to transaction-level purchase data, we observe detailed clickstream data, which enabled us to examine the behavioral mechanisms underlying consumers' responses.

Figure 1 **Cart Value Distributions with CFS (Panel A) and without CFS (Panel B) policies.**



Notes: All completed carts are observed between January 2023 and October 2023. Panel A shows cart values for metropolitan regions under a CFS policy with an $80 threshold, while Panel B shows cart values for non-metropolitan regions without a CFS policy.

To answer the first research question, we document several novel findings using the DiD identification strategy. Following the threshold increase, consumer spending on the platform declined by 19.4%, driven primarily by a 19.0% reduction in average cart value and, to a lesser extent, a 3.1% decrease in purchase frequency. The composition of consumers' baskets also shifted: consumers purchased 7.3% fewer distinct products, switched to lower-priced items, with unit prices declining by 11.5%, and reduced the share of spending allocated to popular products by 3.3%. These effects exhibit substantial heterogeneity, with larger spending reductions among consumers with greater access to alternative grocery channels, higher income, or smaller household sizes.

To answer the second research question, we identify two mechanisms underlying these responses. The first mechanism stems from the combination of reference dependence and goal gradient theory, which we refer to as the reference point effect. CFS policies motivate top-up behavior only when the threshold is perceived as attainable. Consumers who hold the original threshold as a salient reference point may perceive the new, higher threshold as farther away, thereby reducing the activation of top-up behavior. The second mechanism relates to heterogeneity in search effort. Once consumers decide to pursue free shipping, they must search for additional items to reach the threshold. A higher threshold increases search effort, disproportionately discouraging consumers who find it harder to identify suitable items, resulting in smaller baskets or abandoned purchases.

To close the loop and to answer the third research question, we develop a stylized analytical model to study the implications of our empirical findings for optimal free-shipping threshold design. The model integrates both cost-side and demand-side forces that govern the profitability of adjusting the CFS threshold. We first characterize the platform's optimal threshold policy and show that it exhibits a *bang–bang* structure: it is optimal to increase the CFS threshold only when the associated observed average cart value lies within a feasible range, and optimal not to increase it otherwise. This result aligns closely with our empirical evidence, which indicates that consumers' top-up behavior only operates within a certain range. Once the threshold exceeds this range, the resulting loss in demand outweighs the cost savings from reduced shipping subsidies. We conduct a series of robustness analyses by extending the stylized model to incorporate consumer heterogeneity with respect to the distribution of cart values, the reference point effect, and heterogeneous search effort. We show that the *bang–bang* nature of the optimal threshold policy remains robust under these richer and more realistic environments, which reinforces the generalizability of our analytical insights.

Our findings yield several managerial implications. First, managers often rely on observed average cart values under the existing CFS policy to justify higher thresholds (e.g., Shopify 2025, USPS Delivery 2025). We show that this metric can be misleading, as it reflects endogenous top-up behavior rather than the underlying consumer demand. A higher observed average cart value may, counterintuitively, indicate that the threshold should not be increased. Second, platforms must explicitly account for behavioral mechanisms, such as the reference point effect and search effort, when adjusting thresholds. Ignoring these mechanisms can lead to an overestimation of the profitability of higher thresholds. Third, we propose a gradual rollout strategy that exposes higher thresholds only to consumers who have not shopped recently and are therefore less affected by the reference point effect. We show that such a strategy substantially mitigates the negative demand impact of threshold increases.

The remainder of the paper is organized as follows. § 2 reviews the related literature. § 3 describes the research context and data. § 4 presents the treatment effects of the threshold increase. § 5 investigates the underlying mechanisms. § 6 discusses managerial implications. § 7 reports robustness checks, and § 8 concludes.

2. Literature Review

Our work is related to three streams of literature: (1) the design of shipping policies, (2) the demand-side consequences of shipping service characteristics, and (3) the demand-side consequences of shipping fees and policies.

The first stream examines the design of shipping policies. Prior studies have primarily focused on the joint optimization of pricing and margins with free shipping policies. Early works by Leng and Becerril-Arreola (2010) analyzed the interaction between profit margin decisions in marketing and threshold-setting decisions in operations. Subsequent extensions incorporated inventory considerations (Becerril-Arreola et al. 2013) and competitive dynamics, with Shao (2017) examining equilibrium pricing and order quantities under free versus paid shipping policies in competitive settings. More recently, Cachon et al. (2018) developed a data-driven analytical model to evaluate the profitability of CFS and identify optimal thresholds, while Li et al. (2023) derived analytical solutions for the joint optimization of flat-rate shipping fees, free-shipping thresholds, and profit margins.

These analytical studies provide valuable insights into threshold design. However, they typically model consumer response to thresholds using static demand functions that do not capture behavioral mechanisms. Moreover, they focus on setting an initial threshold rather than adjusting an existing one, a distinction that proves critical when consumers form reference points based on their prior shopping experience. Our analytical model in § 6 explicitly incorporates these behavioral mechanisms and examines how optimal thresholds depend on the observed cart values and the presence of the reference point effect.

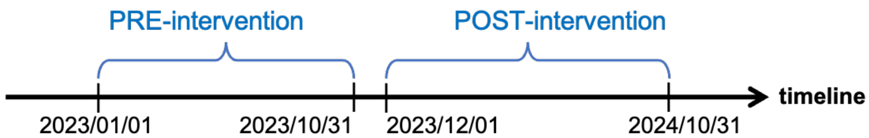
The second stream of research focuses on the demand-side consequences of shipping service characteristics. A growing empirical literature emphasizes how different shipping attributes, such as speed, convenience, and quality, affect consumer behavior. Faster shipping due to proximity to distribution centers increases both online and offline sales (Fisher et al. 2019), while improvements in shipping speed also increase purchase frequency, albeit with slightly smaller basket sizes (Hu et al. 2024). Beyond speed, consumers value precision and reliability in delivery (Amorim et al. 2024), and promises of faster delivery can not only increase sales and profits but also increase return rates and reduce long-term retention (Cui et al. 2024). Convenience also matters, as home delivery options substantially boost sales and spending (Lu et al. 2025). Finally, higher-quality shipping services can increase sales but simultaneously reduce sales dispersion across products (Cui et al. 2020). Our study complements this literature by shifting focus from non-price shipping service attributes to the design and adjustment of price-based shipping policies. Specifically, we examine how dynamic adjustments to free-shipping thresholds affect consumer purchasing.

The third stream examines the demand-side consequences of shipping fees and pricing policies. Early studies have shown that online consumers are highly sensitive to shipping fees, which represent a major source of friction in e-commerce (Brynjolfsson and Smith 2000, Gümüş et al. 2013). Subsequent research has demonstrated that shipping fee structures also affect consumers. For instance, site-wide free shipping promotions increase spending on items that are difficult to evaluate online, expanding their share in the basket and leading to higher return rates (Shehu et al. 2020). In omnichannel retailing, flat-rate shipping (versus tiered rates) shifts purchases from the online channel to offline stores (Kanuri et al. 2025).

A special focus within this literature examines CFS policies. Notably, it has been shown that relative to fixed-fee shipping, CFS leads consumers to purchase less frequently but with larger basket sizes (Lewis et al. 2006). This shift is primarily driven by the reduction in medium-sized orders, as consumers strategically pad orders to meet the free-shipping threshold (Lewis 2006). This "order padding" behavior has been further validated using structural models (Cachon et al. 2018, Hemmati et al. 2021). In addition, consumers are more likely to pad their orders on the PC channel than on the mobile channel (Jin et al. 2023). Beyond changes in basket size, prior research highlights that CFS also shapes consumers' perceptions of fairness and their strategic responses to CFS policies. For instance, consumers evaluate an order less positively when its value is below the CFS threshold, as they consider the shipping fee as a profit generator rather than a cost of doing business (Koukova et al. 2012). Consumers strategically take advantage of CFS by adding unwanted items to qualify for free shipping, then returning them when return processes are favorable (Hemmati et al. 2021, Kabra et al. 2024). Furthermore, Chen and Ngwe (2018) emphasize that consumers cannot meet the minimum spending requirement for free shipping without overshooting. This occurs because consumers face significant search effort when trying to identify products that will help them just meet the threshold.

Despite the prior work on CFS policies, empirical evidence on how adjustments to an existing CFS threshold affect consumer behavior remains scarce. Prior studies typically examine the adoption of CFS versus alternative fee structures, but retailers often dynamically revise thresholds in response to changing cost conditions (Nigrelli 2025). While existing research demonstrates that consumers engage in top-up behavior to meet a given threshold, it remains unclear whether and how consumers adjust their behavior when that threshold increases. Understanding these behavioral consequences is critical, as raising thresholds may unintentionally undermine the incentive structure that makes CFS effective. Our study addresses this gap by providing the first causal evidence on the impact of raising CFS thresholds, identifying the behavioral mechanisms underlying consumer responses, and deriving implications for optimal threshold design.

Figure 2 **Timeline of Our Data Samples.**



3. Research Context and Data

3.1. Research Context

To examine consumer responses to the CFS threshold increase, we leverage a policy change implemented by a North American online grocery platform. Founded in 2019, the platform has grown rapidly, partnering with more than 1,000 suppliers to serve customers across 450 locations and offering a diverse range of 11,000 products across nearly 400 subcategories. Notably, the platform applies different shipping policies based on consumers' location. Consumers residing in metropolitan regions are eligible for the CFS policy, whereas others are subject to standard paid shipping. At checkout, consumers eligible for the CFS policy are shown not only the shipping surcharge based on their current cart value but also the additional amount they need to spend to qualify for free shipping. In contrast, consumers outside the CFS-entitled regions are shown only the applicable shipping surcharge.

The free shipping threshold under the CFS policy was initially set as \$80, and raised to \$100 in November 2023 for all consumers in the CFS-entitled regions. Moreover, the adjustment was implemented without prior announcement, promotional campaigns, or any concurrent policy change. This policy shift provides a quasi-experiment setting, allowing us to isolate and identify the behavioral impact of the threshold increase without confounding influences from other marketing interventions.

3.2. Data

We obtained consumer-level *transaction data* and *clickstream data* from the collaborating platform, covering the period from January 1, 2023, to October 31, 2024. As shown in Figure 2, we divide the sample into three distinct phases: (1) the pre-intervention period (January 2023 – October 2023), (2) the intervention period (November 2023), during which the platform implemented and rolled out the new threshold, and (3) the post-intervention period (December 2023 - October 2024). We restrict our primary analysis to the pre- and post-intervention periods and focus on consumers who placed at least one order in both periods. This restriction, commonly employed in consumer-level analyses (e.g., Guo and Liu 2023, Lu et al. 2025), allows for meaningful within-consumer comparisons of pre- and post-intervention behavior.

We further exclude three types of users likely to bias the estimates: membership users, third-party resellers, and bot-like accounts. Membership users who pay a monthly premium receive free shipping regardless of cart value and are therefore unaffected by the CFS threshold policy. Third-party resellers who place large, frequent orders regardless of shipping policies are unlikely to respond to the threshold change. Bot-like users, identified by high-volume browsing behavior over long, continuous periods without corresponding transactions, likely do not represent real shopping activity. Together, these excluded users account for about 3.3% of the initial sample. After filtering, the final sample comprises 261 consumers (132 from treated regions and 129 from control regions) who placed 1025 orders during the study period. To examine monthly purchase behavior, we construct an imbalanced panel dataset with 5,148 consumer-month observations, where each observation corresponds to a specific consumer in a given month. The panel is imbalanced because not all consumers were active at the beginning of the observation window.

3.2.1. Transaction data The transaction data provides detailed records of each completed order. Each record includes the consumer ID, the shipping address ZIP code, the product ID, timestamps for when each product was added to the cart, the quantity and unit price for each product, and the payment timestamp. Additionally, it also contains information on the pre-intervention popularity (if any) and the retailer's fulfillment costs.

To study product heterogeneity, we classify products by their popularity before the intervention. Specifically, we define popular products as those that collectively account for the top 80% of total units sold during the pre-intervention period. The remaining products are categorized as long-tail products, which together account for the bottom 20% of total units sold.

3.2.2. Clickstream data We obtained detailed consumer clickstream data that captures browsing behavior. Each log entry includes a timestamp, the consumer ID, and the ID of the viewed product.

Combining the transaction and clickstream data, we construct session-based consumer browsing activity. A session is defined as a continuous browsing period initiated when a consumer enters the website and terminated either by a transaction, a period of inactivity (e.g., 30 mins), or an exit (Cooley et al. 1999). Within each session, consumers may perform a variety of observable actions, including viewing product detail pages, adding items to the cart, and proceeding to checkout.

Table 1 Variable statistics

Variables	Observations	Mean	Standard deviation	Min	Max
$Spending_{it}$	5148	29.37	77.57	0	896.67
$AvgCartValue_{it}$	5148	26.87	69.17	0	636.36
$Transactions_{it}$	5148	0.20	0.45	0	5
$Variety_{it}$	5148	1.14	3.30	0	38
$AvgPrice_{it}$	5148	5.56	21.38	0	299
$PopShare_{it}$	5148	0.12	0.29	0	1
$NetRev_{it}$	5148	28.29	75.05	0	896.67

3.2.3. Variables Following prior research (e.g., Guo and Liu 2023), we define two sets of outcome variables that capture (1) how much a consumer purchases and (2) what a consumer purchases. Table 1 summarizes the outcome variables used in the analysis along with their descriptive statistics. The unit of analysis in this study is the consumer-month level.

To measure how much a consumer purchases, we use consumer monthly spending (*Spending*), defined as the total value of completed orders in a given month. Since total spending can be decomposed into the number of transactions and average cart value, we also include *Transactions*, defined as the number of completed orders in a month (a measure of purchase frequency), and *AvgCartValue*, defined as the average cart value per order in that month.

To examine what a consumer purchases, we consider several variables related to cart composition. *Variety* captures the number of distinct products a consumer purchases in a month and serves as a proxy for the variety of purchases. *AvgPrice* measures the average unit price of the total purchased items in a month, with higher values indicating a tendency toward more expensive products. *PopShare* captures the share of spending allocated to purchasing popular products, with higher values indicating greater preferences for popular items.

In addition, we also include net consumer revenue (*NetRev*), defined as the consumer's monthly spending plus shipping surcharges paid, minus the retailer's fulfillment costs (Fisher et al. 2019). *NetRev* serves as a proxy for the platform profit.

As shown in Table 1, all outcome variables exhibit substantial right skewness and contain a nontrivial mass at zero. To address these distributional features, we apply a log-plus-one transformation. Specifically, for each outcome variable $y \in \{Spending, AvgCartValue, Transactions, NetRev, Variety, AvgPrice, NetRev\}$ we construct the transformed variable as $\log(y_{it}+1)$, which we use consistently throughout the regression analyses in this paper. The variable *PopShare* is not transformed, as it is a bounded proportion and already lies within a well-defined range.

Table 2 Model-free Evidence: Descriptive Statistics of Mean Value.

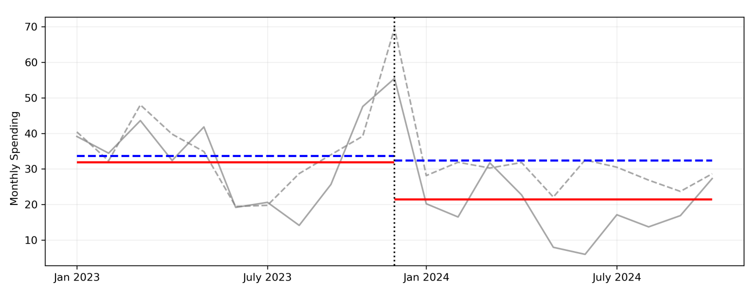
	Control			Treatment		
Variable	**Pre**	**Post**	**Change**	**Pre**	**Post**	**Change**
Spending	36.14	35.55	-1.6%	34.07	21.54	-36.8%
AvgCartValue	33.63	33.01	-1.8%	29.58	19.52	-34.0%
Transactions	0.23	0.20	-14.3%	0.25	0.16	-36.3%
Variety	1.54	1.35	-12.2%	1.37	0.85	-37.7%
AvgPrice	5.90	5.13	-13.1%	6.35	4.72	-25.6%
PopShare	0.14	0.12	-18.3%	0.15	0.09	-38.0%
NetRev	33.98	33.48	-1.5%	33.79	21.32	-36.9%

3.3. Model-free Evidence

We first provide some analyses of the overall impact of CFS threshold increases by comparing changes in purchase behavior between the control and treatment groups before and after the intervention (Table 2). The simple comparison shows that consumers' monthly spending in the treatment group declined more than that in the control group after the CFS threshold increase (-36.8% vs. -1.6%). We find larger reductions in both average cart value (-34.0% vs. -1.8%) and purchase frequency (-36.3% vs. -14.3%). Regarding what consumers purchase, we find that consumers in the treatment group tend to purchase fewer distinct products and prefer cheaper and long-tail products. To visualize the impact, we also show the time series of consumer monthly spending during the observation periods in Figure 3, and include the time series for other outcome variables in the Online Appendix EC.1.

These model-free statistics provide preliminary evidence that raising the CFS threshold adversely affects consumer behavior, as measured by all outcome variables, in the treatment group relative to the control group. We further investigate these patterns in our empirical analysis by accounting for both observed and unobserved factors not captured in the above analysis.

Figure 3 Time Series of Consumer Monthly Spending During the Observation Period



Notes: The grey dashed line represents the time series of the average monthly spending for the control group, and the grey solid line represents the same for the treatment group. The vertical dotted line marks the beginning of the post-intervention period. The blue-dashed horizontal lines indicate the pre- and post-intervention averages for the control group, and the red-solid horizontal lines indicate those for the treatment group.

4. Results

We employ a DiD approach with two-way fixed effects to estimate the causal impact on consumer purchase behavior. Our identification strategy relies on comparing treated and control consumers before and after the policy change, while controlling for time-invariant consumer heterogeneity and common temporal shocks. Specifically, we estimate the following model specification:

$$y_{it} = \alpha_0 + \alpha_1 \cdot Treated_i \times Post_t + \mu_i + \delta_t + \varepsilon_{it} \tag{1}$$

where y_{it} is one of the outcome variables described in Section 3.2.3 for consumer i in year-month t. $Treated_i$ is a binary variable equal to one if consumer i resides in a CFS-entitled region, and zero otherwise. $Post_t$ is a binary variable equal to one if month t is in the post-intervention period, and zero if in the pre-intervention period. The key explanatory variable is the interaction term $Treated_i \times Post_t$, and its coefficient α_1 captures the average treatment effect on the treated. The model includes consumer fixed effects μ_i to control for time-invariant heterogeneity across individuals, and year-month fixed effects δ_t to account for common time trends and seasonal fluctuations.

4.1. Impact on Consumer Spending

Table 3 shows the regression results. Specifically, Column (1) reports the treatment effect on consumer monthly spending. The coefficient indicates that consumers in the treatment group substantially reduced their spending following the increase in the threshold. The coefficient suggests a 19.4% decline ($e^{-0.216} - 1 = -0.194$) in consumer spending from the treatment group after the intervention. To assess the economic significance, we use the post-intervention monthly spending of \$32.36 per consumer in the control group as the baseline. Relative to this benchmark, the threshold increase corresponds to a \$6.28 reduction in monthly spending per consumer in the treatment group. Column (2) presents the treatment effect on the average cart value per order. The estimated coefficient indicates that the average cart value declined by 19.0% ($e^{-0.211} - 1 = -0.190$)

Table 3 Effects of CFS Threshold Increase on Consumer Spending.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*
$Treated \times Post$	-0.216**	-0.211**	-0.032**
	(0.101)	(0.098)	(0.016)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.03	0.03	0.03

*Notes: Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

in the treatment group relative to the control group. Column (3) shows the treatment effect on the number of completed orders. The coefficient suggests that, following the increase in the threshold, consumers in the treatment group placed approximately 3.1% ($e^{-0.032} - 1 = -0.031$) fewer orders per month compared to those in the control group.

Our empirical results in Table 3 reveal nuanced dynamics in how consumers respond to increases in the CFS threshold. Prior empirical studies (e.g., Cachon et al. 2018, Hemmati et al. 2021, Lewis et al. 2006) have shown that CFS policies are effective at consolidating orders and incentivizing consumers to purchase more items than unconditional or flat-fee shipping policies, resulting in higher cart values and higher overall revenue. However, we demonstrate that when the platform raises the CFS threshold, the incentive structure changes. Most notably, the decline in consumer monthly spending is driven primarily by a reduction in cart value rather than a substantial decline in purchase frequency, suggesting that consumers become less willing to expand their cart size in response to the higher threshold. In other words, the loss of motivation to reach the higher CFS threshold leads consumers to revert to their natural purchasing behavior without a free shipping incentive, ultimately eroding the effectiveness of the CFS incentive.

4.2. Impact on Consumer Shopping Pattern

Regarding the impact on consumer shopping patterns, Column (1) of Table 4 shows that the number of distinct products purchased by consumers in the treatment group decreases after the threshold increase. The coefficient corresponds to a 7.3% reduction in purchase variety ($e^{-0.076} - 1 = -0.073$). This decline suggests that consumers engaged in less exploratory or diversified purchasing behavior, potentially due to their weakened incentive to expand cart size. Column (2) reports the effect on the average price of purchased products. The estimated coefficient indicates that consumers shifted towards lower-priced items, as the average unit price of the cart declines by 11.5% ($e^{-0.122} - 1 = -0.115$). This is possibly because there was no incentive to include higher-value products to reach

Table 4 Effects of CFS Threshold Increase on Consumer Shopping Pattern.

	(1) *Variety*	(2) *AvgPrice*	(3) *PopShare*
$Treated \times Post$	-0.076*	-0.122*	-0.033**
	(0.040)	(0.062)	(0.015)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.02	0.04	0.04

*Notes: Standard errors are clustered at the consumer level. Both Variety and AvgPrice are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

the free shipping threshold. Column (3) presents the effect on the share of popular products in total purchases. The result indicates a 3.3% decline in the proportion of popular items purchased. This decline suggests that in the absence of the motivating force of the CFS benefit, consumers were less likely to add popular products, which are commonly used to top up cart values, to their orders. Instead, the relative share of long-tail products increased, implying that consumers were willing to purchase more niche products even without the incentive of free shipping.

Taken together, the results in both Tables 3 and 4 reveal a consistent pattern: the increase in the CFS threshold not only suppressed how much consumers purchased, but also shifted the composition of what they purchased. Unlike the top-up behavior documented in prior studies, consumers reduced both the average price and the variety of products purchased. Without the incentive to qualify for free shipping, consumers reverted to more utilitarian shopping patterns, focusing on essential and personally valued items, including long-tail products, even if it meant incurring shipping fees.

4.3. Heterogeneous Treatment Effect

We also investigate how the impact varies across consumers. Specifically, we consider three dimensions of a neighborhood, namely grocery alternatives, affluence, and household composition. We use split samples to conduct the analysis (see Online Appendix EC.2 for details).

Grocery alternatives may shape consumer behavior, prompting consumers to switch to other online or offline channels when the shipping policy changes. Consumers with easy access to nearby physical grocery stores face lower switching costs and are less “locked in” to the focal platform, whereas those without nearby alternatives have fewer substitution options and may thus exhibit smaller behavioral changes. Consistent with this reasoning, the regression results in Table EC.1 show that the CFS threshold increase significantly reduced total spending and average cart value among consumers with high local alternatives, while the effects for those with fewer alternatives are statistically insignificant.

Neighborhood affluence may also shape consumers’ responses to increases in the CFS threshold. Consumers in lower-income areas face tighter budget constraints and are likely more sensitive to shipping fees. When the threshold is raised, they may respond more strongly by reducing spending or purchase frequency. In contrast, consumers in higher-income neighborhoods experience weaker budget constraints and are thus less affected. The results in Table EC.2 confirm this reasoning: both total spending and average cart value decline significantly for consumers in low-income neighborhoods, whereas consumers in higher-income areas exhibit negligible changes.

Household composition may further moderate consumers' sensitivity to increases in the CFS threshold. Larger households typically purchase more groceries per order, so their routine baskets often approach or exceed the free-shipping threshold, making them less affected by the policy change. By contrast, consumers in smaller-household neighborhoods place smaller baskets and have less flexibility to scale up expenditures merely to avoid shipping fees, leaving them more exposed to the threshold increase. Consistent with this logic, Table EC.3 shows that consumers in smaller-household neighborhoods exhibit significant declines in total spending and purchase frequency, whereas the effects are statistically insignificant for those in larger-household areas.

5. Mechanisms

We next explore why consumers reduce their spending following the threshold increase. We identify two possible mechanisms. The first mechanism combines reference dependence and goal gradient theory, which we refer to as the reference point effect, and the second mechanism relates to the heterogeneity in search effort.

Our first mechanism builds on the interaction between reference dependence theory and goal gradient theory. Reference dependence theory shows that consumers evaluate outcomes relative to reference points, internal benchmarks formed from past experience, rather than in absolute terms (Kahneman and Tversky 1979, Tereyağoğlu et al. 2018). In CFS settings, consumers who repeatedly shop under a given threshold may internalize that threshold as their reference point for what constitutes a "reasonable" cart value to qualify for free delivery. Once established, this reference point becomes the benchmark against which the new, higher requirement is evaluated.

Goal gradient theory predicts that consumers accelerate effort as they approach a reward, but this effect presumes that consumers have activated the goal in the first place (Choudhary et al. 2021, Kivetz et al. 2006, Lim et al. 2023). Goal activation depends on whether the target is perceived as attainable. When a threshold increases, perceived attainability may shift because the new requirement is evaluated relative to the existing reference point. The interaction between these two theories is critical: reference points shape how distant the new threshold feels, which in turn determines whether consumers activate goal gradient behavior or disengage from pursuing free shipping altogether.

In our setting, the threshold increases from \$80 to \$100. Consumers who recently shopped under the old policy are likely to hold \$80 as a salient reference point. For these consumers, the new threshold may be perceived not simply as a \$100 target but as requiring an additional \$20 beyond what they view as the normal qualification level, a 25% increase from their reference point. This increases the perceived distance to the goal and may weaken incentives to engage in top-up behavior. Instead of adding items to reach \$100, these consumers may revert to purchasing only their intended items and accept the shipping fee, resulting in smaller carts and lower spending. In contrast, consumers whose memory of the prior threshold has faded may evaluate the \$100 requirement more independently of the old \$80 benchmark, making the free-shipping goal more attainable and preserving their motivation to top up their carts. We therefore expect larger spending reductions among consumers who recently shopped under the old threshold, for whom the prior reference point remains most salient.

Our second mechanism recognizes that, even after deciding to engage in topping up, reaching a free-shipping threshold requires consumers to search for and evaluate additional products to add to their carts. This process involves time and cognitive effort, as consumers must identify items that are useful enough to justify purchasing, that help close the remaining gap to the threshold, and that do not cause them to overshoot by a large margin (Chen and Ngwe 2018). When the threshold is raised, the gap consumers must close increases, thereby raising the total search effort required to qualify for free shipping.

Importantly, the search effort required to meet a given threshold varies across consumers. Some consumers are familiar with the platform's product assortment, have flexible preferences, or can quickly navigate the website. For these low-effort consumers, identifying additional items to reach a higher threshold may require only modest additional effort. Other consumers face higher search effort due to limited product familiarity, narrower preferences, or difficulty identifying suitable items. For these consumers, the same increase in the threshold translates into substantially more time and cognitive effort.

When the threshold increases from \$80 to \$100, the added search burden affects all consumers but not equally. Low-effort consumers may still find it worthwhile to search for additional items and continue to engage in top-up behavior. High-effort consumers, however, face a steeper increase in effort, which may render the free-shipping benefit less attractive relative to the cost of pursuing it. As a result, they are more likely to abandon the pursuit of free shipping, complete smaller purchases, or reduce purchase frequency. We therefore expect that spending reductions should be larger among consumers who face higher search effort.

Table 5 **Effects of CFS Threshold Increase: The Role of the Reference Point Effect.**

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	0.073	0.075	0.013
	(0.105)	(0.104)	(0.015)
$Treated \times Post \times Recent$	-0.550***	-0.543***	-0.085***
	(0.192)	(0.187)	(0.030)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.03	0.03	0.03

*Notes: Standard errors are clustered at the consumer-level. All variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

To test our mechanisms, we use the following model specifications:

$$y_{it} = \beta_0 + \beta_1 \cdot Treated_i \times Post_t + \beta_2 \cdot Mechanism_i \times Treated_i \times Post_t + \beta_3 \cdot Mechanism_i \times Post_t + \mu_i + \delta_t + \varepsilon_{it} \tag{2}$$

where $Mechanism_i$ denotes the proxy variable representing one of the two mechanisms discussed above, the coefficient β_1 measures the average treatment effect in the baseline group, and β_2 measures the difference in the treatment effect between high and low levels of the mechanism. Finally, β_3 captures any general time trend difference across consumers with different $Mechanism_i$ values that is unrelated to treatment.

5.1. Reference Point Effect

We operationalize the first mechanism using consumers' recency of purchase before the policy change as a proxy for reference point salience. Recent experiences are more cognitively accessible and salient than distant ones, making them more likely to serve as active reference points. Specifically, we define $Recent_i = 1$ if a consumer's last pre-intervention purchase occurred within three months (the sample median) before the threshold increase, and $Recent_i = 0$ otherwise.

Table 5 reports the estimation results. The coefficient on $Treated \times Post$ is positive, suggesting a goal gradient effect on consumers for whom the old threshold was not salient. However, it is not statistically significant, indicating that the increase is not strong enough. In contrast, the interaction term $Treated \times Post \times Recent$ is negative and statistically significant across all outcomes. The estimates indicate that, relative to consumers less influenced by the old threshold, recent purchasers reduced their total spending and average cart value by approximately 55% and their purchase frequency by about 8.5% following the policy change. These results are consistent with the reference dependence and goal gradient theory. When the original threshold remains salient, the higher target feels more distant, weakening consumers' motivation to pursue free shipping and leading to larger reductions in spending.

Table 6 Effects of CFS Threshold Increase: The Role of Search Effort.

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.061	-0.056	-0.009
	(0.117)	(0.114)	(0.019)
$Treated \times Post \times HighEffort$	-0.538**	-0.533**	-0.080**
	(0.223)	(0.218)	(0.034)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.03	0.03	0.03

*Notes: Standard errors are clustered at the consumer-level. All variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

5.2. Search Effort

We measure consumers' search effort by the time they spend adding items to their carts before the policy change. Specifically, for each order, we calculate the time difference between (i) the average time spent selecting the last two items and (ii) the average time spent for all items in that order. A positive time difference indicates that the consumer slows down toward the end of the shopping process, suggesting greater difficulty in identifying additional products to reach the threshold, an indicator of higher search effort. We then aggregate this measure at the consumer-level: if, across all pre-intervention orders, a consumer tends to spend more time on the last items, we classify them as having high search effort ($HighEffort_i = 1$); otherwise, as low search effort ($HighEffort_i = 0$).

Table 6 reports the heterogeneous treatment effects by search effort. Despite its insignificance, the coefficient on $Treated \times Post$ is negative across all three outcomes for low-search-effort consumers, suggesting that the threshold increase also imposes additional search effort on this group. In contrast, the interaction term $Treated \times Post \times HighEffort$ is negative and statistically significant for spending, average cart value, and transactions. Specifically, the estimated coefficients imply that, relative to low-effort consumers, high-effort consumers reduced their total spending and average cart value by approximately 54% and their purchase frequency by about 8% following the policy change. These results provide strong evidence consistent with the search effort mechanism. Consumers who face greater difficulty identifying additional items are more discouraged by the higher threshold and cut back their purchases more sharply.

6. Optimal Threshold Design

In this section, we address the last research question: What are the implications for optimal threshold design? We first empirically investigate whether increasing the threshold to $100 benefits the platform's profit or not. We use monthly net consumer revenue (*NetRev*) to proxy for profit, which accounts for reductions in spending and purchase frequency, as well as the revenue gain from additional shipping surcharges.

Table 7 Effects of CFS Threshold Increase on Platform Profit

	(1)
	NetRev
$Treated \times Post$	-0.218**
	0.100
Consumer Fixed Effects	Yes
Yearmonth Fixed Effects	Yes
Observations	5148
Adjusted R^2	0.03

*Notes: Standard errors are clustered at the consumer-level. * p<0.10, ** p<0.05, *** p<0.01.*

We employ the DiD analysis using specification (1) with the log-plus-one transformed *NetRev* as the outcome variable and report the regression results in Table 7. The coefficient is negative and statistically significant, indicating that raising the threshold from \$80 to \$100 reduced the profit by approximately 19.6%. This suggests that the spending reduction following the policy change outweighed the revenue gain from shipping surcharges.

Motivated by this finding, we then examine how to set the free shipping threshold to maximize consumer net revenue at the cart level (*NetCartRev*). We focus on a single-cart framework for two reasons. First, the threshold increase has only a marginal effect on purchase frequency, implying that changes in cart value are the primary driver of revenue outcomes. Second, the single-cart setting allows us to explicitly model both the shipping surcharge paid by the consumer and the fulfillment cost incurred by the retailer, which are central to threshold design.

Our counterfactual analysis proceeds as follows. We first analyze the optimal threshold under the assumption of homogeneous consumers, using the estimated elasticity of cart value with respect to the threshold. We then relax this homogeneity assumption by explicitly accounting for heterogeneity in cart values and incorporating the reference point effect and search effort, thereby allowing us to characterize how the implied optimal threshold changes when these behavioral mechanisms are incorporated. Next, we discuss how the platform can operationalize threshold increases in practice by leveraging the reference point effect.

6.1. The Optimal Thresholds

We assume that the cart value, denoted by $y(\tau)$, responds with a constant elasticity to the free shipping threshold τ locally around the observed policy change. Specifically, we assume a log–log relationship between cart value and the threshold, i.e., $\ln y(\tau) = \ln y(\underline{\tau}) + \hat{c}\, \ln\big(\tau/\underline{\tau}\big)$, which implies

$$y(\tau) = y(\underline{\tau}) \cdot \big(\tau/\underline{\tau}\big)^{\hat{c}}. \tag{3}$$

Here, $\underline{\tau}$ marks the original threshold value, $y(\underline{\tau})$ is a parameter representing the observed average cart value at the original threshold, and $\hat{c}$ is the elasticity parameter calibrated from the observed DiD estimation when the threshold increases from $\underline{\tau}$ to $\overline{\tau}$.

The platform incurs a per-order fulfillment cost of C dollars and may charge the consumer a shipping surcharge of $S(y,\tau) \geq 0$ dollars, which depends on the realized cart value y and the free-shipping threshold τ. We model the shipping surcharge as

$$S(y,\tau) = \begin{cases} C - ky & y < \tau \\ 0 & y \geq \tau \end{cases}$$

where $k \in [0, C/\tau]$ governs how the surcharge varies with cart value. When $k = 0$, the policy corresponds to a standard CFS policy with a fixed surcharge below the threshold. When $k > 0$, the surcharge declines with cart value, capturing a partially discounted CFS policy. $k \leq C/\tau$ guarantees that the surcharge is still positive before the cart value hits the threshold. The platform chooses the free-shipping threshold within the range $[\underline{\tau}, \overline{\tau}]$ to maximize its expected per-order net revenue:

$$\tau^* \in \arg\max_{\underline{\tau} \leq \tau \leq \overline{\tau}} \; y(\tau) + S(y(\tau), \tau) - C$$

Consequently, we have the following result to characterize the optimal threshold:

PROPOSITION 1 **(Optimal Free Shipping Thresholds)**. *Let $c < 0$. Define $\hat{\tau}(a) = \left(a \cdot \underline{\tau}^{-c}\right)^{\frac{1}{1-c}}$. Then, there exists a cutoff $a^* > 0$, implicitly defined by $(1-k)\hat{\tau}(a^*) = a^* - C$, such that the optimal free-shipping threshold satisfies $\tau^* = \hat{\tau}(a)$ if and only if $a \in [\underline{\tau}, a^*]$ and $\hat{\tau}(a) \leq \overline{\tau}$; otherwise, $\tau^* = \underline{\tau}$.*

The proof can be seen in Online Appendix EC.3.1. The parameter a can be interpreted as the observed average cart value at the baseline free-shipping threshold $\underline{\tau}$. Proposition 1 shows that the platform should increase the free-shipping threshold only when this original demand level is sufficiently close to the original threshold.

The key intuition is that the platform's profit is strictly decreasing in the threshold τ within both regimes, whether orders qualify for free shipping or incur a shipping surcharge. As a result, the optimal policy must lie at one of two points: the minimum allowable threshold $\underline{\tau}$, or the kink $\hat{\tau}(a)$ at which the representative order is just excluded from free shipping. Raising the threshold beyond this kink only reduces demand without generating additional shipping revenue. The cutoff a^* is determined by the indifference condition $(1-k)\hat{\tau}(a^*) = a^* - C$, which equates the profit from extracting shipping revenue at the kink to the profit from offering free shipping at the original threshold. When $a \leq a^*$ and the kink is feasible, increasing the threshold to $\hat{\tau}(a)$ allows the platform to monetize shipping without inducing a large reduction in the cart value. When original cart values are higher, however, the loss in demand from raising the threshold outweighs the shipping revenue that can be extracted, making it optimal to keep the threshold at its minimum level $\underline{\tau}$.

To operationalize the model, we calibrate the elasticity parameter using the log change in average cart value reported in Table 3. Specifically, the estimated elasticity is $\hat{c} = \frac{-0.211}{\log(100/80)} = -0.946$. We then numerically search for the optimal threshold. We consider original average cart values $y(80)$ ranging from \$70 to \$130, covering both thresholds and the observed average cart value during the pre-intervention period. Under the log–log assumption, we set $\underline{\tau} = 80$ and $\overline{\tau} = 100$. We fix the per-order fulfillment cost at $C = 16$ and set $k = 0$. With the above parameters, we can obtain $a^* \simeq 110.41$.

Figure 4 **Optimal Threshold Values and Revenue Improvement with Various Base Cart Values**

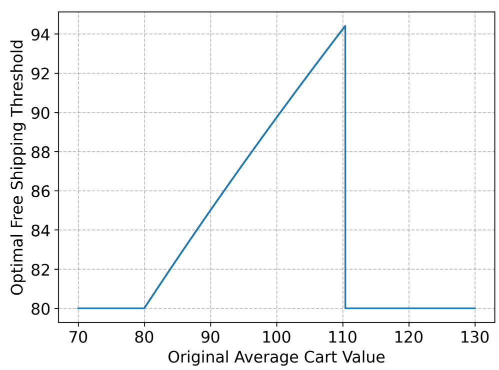


Figure 4 presents the counterfactual results. It follows Proposition 1 and shows a pronounced nonlinear relationship between the optimal thresholds and the original average cart value $y(80)$. Specifically, when $y(80)$ lies between \$80 and \$110.41, a moderate increase in the threshold becomes optimal. In this range, although a slightly higher threshold discourages consumer spending, the platform benefits from reducing subsidies for free shipping and collecting additional shipping surcharges. In contrast, when $y(80)$ is sufficiently high (larger than 110.41), it is optimal not to raise the threshold at all. The reason is that the cart value reduction dominates, and the loss cannot be offset by additional shipping surcharges. For our partner platform, whose pre-treatment average cart value was \$127.17, it should not increase the threshold at all.

We also vary the discounting factor k and report additional insights in Online Appendix EC.3.2. Without accounting for the potential impact of shipping discounts on cart value, our results show that offering larger discounts on shipping surcharges makes suboptimal threshold choices more costly. Given that determining optimal thresholds is far simpler, our results suggest the platform should not offer partially discounted CFS policies.

6.1.1. Accounting for cart value heterogeneity. We extend the preceding analysis to a stochastic threshold optimization problem by allowing cart values at the original free-shipping threshold, $y(\underline{\tau})$, to be heterogeneous across consumers. Specifically, we assume that baseline cart values follow a uniform distribution and study how the platform optimally chooses the free-shipping threshold to maximize expected per-cart net revenue.

Although stylized, this uniform specification delivers a clear economic insight: the optimal free-shipping threshold depends not only on the average but also on the dispersion of cart values. Increasing the threshold τ raises the qualification cutoff and shifts a subset of orders from the free-shipping regime into the surcharge regime. Unlike the deterministic case, the marginal benefits therefore depend on the amount of probability mass near this cutoff, that is, on the distribution of baseline cart values rather than on their mean alone. In particular, holding the average cart value fixed, greater dispersion reduces the density of orders around the cutoff and weakens the incentive to raise the free-shipping threshold. We formalize these results in Online Appendix EC.3.3, where we derive the expected profit function and characterize the optimal threshold in Proposition EC.1.

6.1.2. Accounting for Reference Point Effect and Search Effort. We also extend the counterfactual analysis by explicitly accounting for consumer heterogeneity in the reference point effect and search effort. We decompose the aggregate response to the threshold increase into segment-specific components associated with these two mechanisms, and then re-evaluate the optimal free-shipping threshold using the same optimization framework.

For consumers in the baseline segment who are neither recently exposed to the old threshold nor subject to high search effort, the estimated elasticity $\hat{c}_{baseline}$ is positive. This implies that, in the absence of the behavioral mechanisms, raising the threshold continues to activate goal-gradient behavior, leading these consumers to respond to a higher target by expanding their baskets rather than reducing them. For this group, increasing the free-shipping threshold is therefore revenue-enhancing, and the threshold is better set to \$100, regardless of the original average cart value. In contrast, consumers subject to a salient reference point or high search effort exhibit sharply negative elasticities. For these segments, even modest threshold increases yield substantial reductions in cart value, rendering them unprofitable across a wide range of baseline cart values. Incorporating the reference point effect and search effort, therefore, substantially narrows the set of demand conditions under which raising the free-shipping threshold is optimal and places tighter bounds on feasible threshold levels. We provide further details in Online Appendix EC.3.4.

Overall, these results highlight the importance of accounting for the discovered mechanisms when adjusting free-shipping thresholds. Ignoring the reference point effect and search effort can lead to predictions and policy choices that are qualitatively incorrect.

6.2. Gradually Implementing Threshold Increases

The previous sections discussed how to set optimal thresholds. In this section, we provide an implementable strategy for platforms that must raise thresholds. In such settings, the relevant managerial question is not only how much to increase the threshold, but also which consumers should be exposed to the higher requirement. Our previous analysis on the reference point effect suggests that platforms can raise thresholds more effectively by managing exposure. Specifically, platforms may gradually roll out a higher threshold, exposing it only to consumers who are less likely to treat the original threshold as a salient reference point. Such a strategy may mitigate revenue losses from the reference point effect while still helping the platform address rising fulfillment costs.

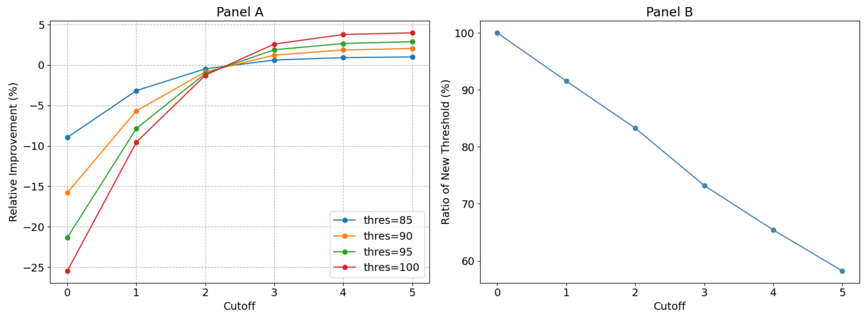
To evaluate the potential benefits of gradual implementation via selective exposure, we use a Monte Carlo simulation with bootstrapped data generation. We characterized consumers by the number of orders completed before the policy change and estimated the empirical distribution. Then, we randomly assign each simulated consumer a type by sampling from the empirical distribution. After that, we generate a sequence of purchase events over time using a bootstrapping procedure. Specifically, inter-purchase times and baseline cart values are resampled from pre-treatment transactions for consumers of the same type. This approach ensures that the simulated purchase process matches observed heterogeneity in purchase frequency and spending behavior.

Under the gradual rollout policy, the platform displays a higher threshold only to consumers whose current inter-purchase time exceeds a cutoff value. Once a consumer is exposed, the higher threshold remains in effect for all subsequent purchases. Upon exposure, the consumer's realized cart value is adjusted to reflect the reference point effect using (3), with the parameter $\hat{c}$ determined as follows:

$$\hat{c} = \frac{\hat{\beta}_1 + \hat{\beta}_2 \log(\text{InterTime})}{\log(100/80)}$$

where InterTime denotes the consumer's current inter-purchase time, $\hat{\beta}_1$ captures the baseline impact of an increased threshold, and $\hat{\beta}_2$ captures the differential impact moderated by the strength

Figure 5 **Time Series of Consumer Monthly Spending During the Observation Period**



Notes: The final adoption rate is independent of the threshold values. Therefore, the right panel shows the adoption rate as a function of the cutoff value with the threshold of $100.

of the reference point effect. Using a Triple DiD analysis with continuous inter-purchase time in Online Appendix EC.3.5, we obtain $\hat{\beta}_1 = -0.674$ and $\hat{\beta}_2 = 0.380$.

We track two key outcomes. First, we compute the platform's profit under a gradual rollout and compare it with that under the original threshold. We use the relative profit improvement to quantify potential profitability gains. Second, we monitor the proportion of consumers who see the new threshold at the end of the simulation horizon, which captures the speed of adoption of the higher threshold under the gradual rollout policy. We set the number of consumers to 131 and the simulation horizon to 12 months. We vary the threshold from $80 to $100 in $5 increments and vary the cutoff from 0 to 5 months. Note that a cutoff of 0 corresponds to immediately exposing the higher threshold to all consumers, mirroring the policy implemented by our partner platform.

The simulation results are shown in Figure 5. The left panel shows that gradual implementation can substantially mitigate or even fully eliminate the profit losses associated with higher thresholds. By selectively exposing the higher threshold to consumers who are less affected by the reference point, the platform dampens the negative response that arises under uniform implementation. In particular, when the platform restricts exposure to consumers with inter-purchase times exceeding 3 months, the gradual rollout policy yields higher profits. Beyond this point, further increases in the cutoff yield additional profit gains, though these gains exhibit diminishing marginal returns. In addition, if an appropriate cutoff value is selected, the new threshold is less important, since gradual rollout policies with different thresholds yield similar outcomes.

In contrast, the right panel highlights the corresponding trade-off in adoption speed. More selective rollouts necessarily slow the diffusion of the higher threshold across the consumer base. As the cutoff increases, a larger fraction of consumers continues to see the original threshold at the end of the simulation horizon. For example, with a 5-month cutoff, only 56% of consumers are subject to the new threshold after one year. Thus, while higher cutoffs improve short-run profit performance by limiting exposure among consumers recently exposed to the old threshold, they also delay the full adoption of the new threshold and complicate platform operations. Taken together, moderate cutoffs, such as exposing the higher threshold only to consumers with inter-purchase times longer than 2 or 3 months, can deliver most of the profit benefits of a gradual rollout while avoiding excessively slow adoption.

7. Robustness Checks

We conduct a series of robustness checks for our main results. We start with testing the parallel trends assumption and conducting placebo tests. We also address non-random assignment using coarsened exact matching (CEM) to reduce imbalance between the treatment and control groups. To mitigate selection bias arising from restricting the sample to consumers who made at least one purchase before and after the intervention, we use an inverse probability weighting (IPW) estimator and an aggregation-level analysis.

7.1. Parallel Trends Assumption

We examine the parallel trends assumption in DiD analysis, i.e., that the treatment and control groups follow parallel trends in the outcome variables during the pre-intervention period. To do so, we conduct a pre-trends test and examine whether the differences in the trends of the dependent variables between the treatment and control groups change linearly with time during the pre-intervention period. We use the following model specifications for this test:

$$y_{it} = \tau_0 + \tau_1 \cdot Treated_i \times Month_t + \mu_i + \delta_t + \varepsilon_{it} \tag{4}$$

where $Month_t$ is a categorical variable that represents the month in the pre-intervention period. This specification allows a different coefficient for each time point in the pre-intervention period. It does not impose a linear assumption about the trends in the differences between the treatment and control groups. We present the estimation results in Table EC.7 in the Online Appendix EC.4.1. We find that, across all outcome variables in Section 3.2.3, the interaction term coefficients are *insignificant*, suggesting no statistically significant differences in the trends of the dependent variables between the treatment and control groups during the pre-intervention period.

7.2. Placebo Tests

We also conduct a placebo test to rule out the possibility that the estimated effect is spuriously driven by factors other than an increase in the CFS threshold. Specifically, we first introduce a "fake CFS threshold increase" during the pre-intervention period and estimate its impact on the dependent variables using the pre-intervention period data. We set our fake increase to be on June 1, 2023, which is the middle of the pre-intervention period in our sample. We use a specification that replaces the variable $Post_t$ in Specification (1) by the variable $PlaceboPost_t$, which is a dummy variable that equals one if time t is after the fake threshold increase and zero otherwise. Next, we introduce a "fake treatment group" among consumers in the actual control group and estimate the treatment effect on the dependent variables using the control group data. We randomly select 64 consumers from the control group and assign them to the fake treatment group, and the remaining 65 consumers stay in the control group. We use a specification that replaces the variable $Treated_i$ in Specification (1) with $PlaceboTreated_i$, a dummy variable that equals one if consumer i belongs to the placebo treatment group and zero otherwise. The estimation results are shown in Table EC.8 in the Online Appendix EC.4.2. We find no statistically significant effects of the placebo treatment on the dependent variables, suggesting that the estimated effects of the CFS threshold increase are not spurious.

7.3. Non-random Assignment

Because consumers are not randomly assigned to the treatment and control groups, their shopping behaviors may differ systematically, potentially violating the parallel trends assumption required for the DiD identification. However, as shown in Table EC.9, consumers' browsing and purchasing behaviors are largely statistically indistinguishable between the groups prior to the intervention. Nevertheless, to further improve covariate balance and mitigate concerns arising from non-random assignment, we implement coarsened exact matching (CEM) (Iacus et al. 2009). CEM is a nonparametric matching method that allows researchers to pre-specify the level of acceptable imbalance between treated and control units and has been commonly used in DiD applications (e.g., He et al. 2023, Kumar et al. 2019, Mejia et al. 2021).

Specifically, we match consumers in the treatment group to those in the control group based on pre-intervention browsing and purchasing characteristics. We include a detailed matching process in the Online Appendix EC.4.3. After matching, we drop 19 consumers from the treatment group and 25 consumers from the control group. We then estimate Specification (1) on the matched sample, incorporating the CEM-derived weights as analytical weights in the regression. Table EC.11 reports the treatment effect estimates based on the weighted DiD analysis. The results are consistent with our main findings and provide additional support for the robustness of our conclusions.

7.4. Impacts with Expanded Samples

A potential concern with our main analysis is the possibility of selection bias arising from restricting the sample to consumers who made at least one purchase both before and after the intervention. This restriction may exclude consumers who stopped purchasing altogether after the CFS threshold increase, i.e., those who churned from the platform. If the policy change discouraged these consumers from returning, the resulting attrition would lead to an under-representation of highly sensitive consumers in the post-intervention period. As a result, our estimates may understate the true negative impact of the intervention.

However, we note that our main estimates are likely conservative. This is because including intervention-induced churned consumers, who would contribute zero post-intervention purchases, would further magnify the estimated treatment effect. In other words, the observed declines in spending and cart size among retained users represent a lower bound on the overall behavioral response to the policy change.

Despite that, to formally assess and mitigate this potential bias, we conduct a series of robustness checks using the sample of consumers who placed at least one order before the intervention. Specifically, we first adopt an inverse probability weighting (IPW) estimator (Robins et al. 1994). In addition, we aggregate the data to the borough-month level to examine treatment effects at a lower granularity that includes both active and inactive consumers, which is widely adopted in the literature to investigate the impact of a policy change (e.g., Cui et al. 2020, Fisher et al. 2019).

7.4.1. IPW estimator. The key idea behind IPW is to reweight the observed sample to resemble the full population that includes both retained and churned consumers (Robins et al. 1994). In our context, IPW accounts for the possibility that some consumers, especially those negatively affected by the policy change, may have stopped purchasing altogether after the intervention.

Specifically, we first estimate the probability that a consumer remains active after the intervention (i.e., places at least one order in the post-intervention period) based on their pre-intervention aggregation-level characteristic vector Z_i. We construct stabilized IPW weights (Robins et al. 2000) for each consumer as:

$$w_i = \frac{\mathbb{P}(Survive_i = 1)}{\mathbb{P}(Survive_i = 1 \mid Z_i)} \tag{5}$$

where w_i is the stabilized IPW weight for consumer i. $Survive_i = 1$ represents that consumer i places at least one order in the post-intervention period and zero otherwise. The numerator stabilizes the weights and reduces their variance by multiplying the estimated inverse probability by the overall retention probability. We provide detailed information about how to obtain the IPW weights in the Online Appendix EC.4.4.

Using these stabilized weights, we re-estimate our DiD model in Specification (1) to recover treatment effects that are representative of the broader population, including consumers who have not placed any orders during the post-intervention period. Table EC.13 summarizes the estimation results. The estimated effects on all outcome variables remain negative and statistically significant, and their magnitudes are comparable to those reported in Section 4.1.

7.4.2. Aggregation-level analysis. We further conduct an aggregated analysis using a balanced panel at the borough-month level. This approach allows us to assess the robustness of our main findings at a coarser geographic resolution, including both active and inactive consumers. Specifically, we consider outcome variables that are closely aligned with those defined in Section 3.2.3. We defer the detailed analysis to Online Appendix EC.4.5 and only highlight the results as follows.

Employing an aggregated-level DiD analysis, we show that the threshold increase led to a meaningful reduction in overall spending and average cart size at the borough level. The effect on transaction volume is negative but not statistically significant, which reflects the finding that transactions contribute marginally in our main results. In addition, we find significant declines in purchase variety, average product price, and the share of popular products purchased. These findings closely mirror our consumer-level results and further reinforce the conclusion that the increase in the CFS threshold discouraged consumer engagement, both in terms of how much they bought and what they chose to buy.

8. Conclusion

In this paper, we empirically investigate the causal effect of increasing the CFS threshold on consumer purchase behavior using a DiD approach at the consumer-month level. We find that an increase in the CFS threshold leads to a significant decline in consumer monthly spending, primarily driven by a reduction in average cart value, with a marginal decline in consumer purchase frequency. We also document substantial heterogeneity in these responses: consumers differ in their sensitivity to the threshold change depending on their access to alternative grocery channels, their grocery budgets, and their household needs. Furthermore, our analysis of net revenue indicates that setting a higher threshold does not alleviate the challenge of rising fulfillment costs for online retailers and may, in fact, reduce profits.

To unpack the underlying drivers, we identify two mechanisms: the reference point effect and the search effort. First, consumers who hold the original threshold as a salient reference point perceive the new requirement as less attainable. This weakened sense of goal progress reduces their motivation to top up, leading to sharper declines in spending and average cart value. Second, consumers with high search effort face disproportionate difficulty when trying to meet the higher threshold. As a result, they are less likely to add extra items and more likely to abandon or reduce their basket size. Together, these mechanisms reveal how higher thresholds can erode the incentive power of CFS by increasing cognitive effort and diminishing perceived attainability.

These findings offer important implications for managers of online retail platforms. First, the observed average cart value under the original CFS policy could be a misleading signal for deciding whether a threshold increase is beneficial. The observed average cart value reflects consumers' endogenous top-up behavior in response to the existing threshold rather than their underlying purchase needs. When the threshold increases, these behavioral responses change, weakening top-up incentives and potentially leading to lower realized cart values that cannot be compensated by shipping surcharges. Therefore, a higher threshold can be profitable only when the existing average cart value lies sufficiently close to the current threshold.

Second, platforms must explicitly account for consumer psychology when adjusting free-shipping thresholds. Our results show that raising the threshold is revenue-enhancing for consumers who are not subject to a salient reference point or high search effort, as top-up behavior continues to operate for this group. In contrast, for consumers subject to a salient reference point or high search effort, even modest threshold increases can substantially reduce cart values and overall spending. For these consumers, threshold increases are profitable only when baseline cart values are relatively low and close to the original threshold. Ignoring these behavioral mechanisms can therefore lead platforms to overestimate the benefits of threshold increases and inadvertently erode profit.

Third, for platforms that must raise their free-shipping thresholds to offset rising fulfillment costs, we propose a gradual rollout strategy that selectively exposes the higher threshold to consumers for whom the original threshold is less salient. We demonstrate that such a policy substantially dampens the negative impact of uniform threshold increases on consumer spending and platform profit. Meanwhile, it still allows the platform to move toward adopting the new threshold.

We conclude by outlining the study's limitations and suggesting directions for future research. First, our study is based on data from a single online grocery platform in North America without a return policy, which may limit the external validity of the findings. Therefore, future research should examine similar threshold adjustments in other retail settings, such as different product types, customer demographics, cultural contexts, or marketplaces with return options, to identify the boundary conditions of our findings. Second, the study focuses on short- to medium-term behavioral shifts following the policy change. It is unclear whether the observed reductions in spending and cart size persist in the long run or whether consumers eventually adapt to the new threshold. Future research can extend the observation period to study persistence, learning, and habit formation. Third, our work examines only two possible underlying mechanisms without explicitly measuring psychological constructs through survey or experimental methods. Future research may complement our field-based approach with surveys, lab experiments, or field experiments to directly investigate the contributions of these two mechanisms and other potential mechanisms.

References

Amorim P, DeHoratius N, Eng-Larsson F, Martins S (2024) Customer preferences for delivery service attributes in attended home delivery. *Management science* 70(11):7559–7578.

Baymard Institute (2025) 49 Cart Abandonment Rate Statistics 2025. URL `https://baymard.com/lists/cart-abandonment-rate`, [Accessed 2025-08-18].

Becerril-Arreola R, Leng M, Parlar M (2013) Online retailers' promotional pricing, free-shipping threshold, and inventory decisions: A simulation-based analysis. *European Journal of Operational Research* 230(2):272–283.

Brandt L (2018) Online-Shopping in the U.S. 2018. URL `https://www.statista.com/study/59056/online-shopping-in-the-us/`, [Accessed 2025-08-18].

Brynjolfsson E, Smith MD (2000) Frictionless Commerce? A Comparison of Internet and Conventional Retailers. *Management Science* 46:563–585, ISSN 0025-1909, 1526-5501, URL `http://dx.doi.org/10.1287/mnsc.46.4.563.12061`.

Cachon GP, Gallino S, Xu J (2018) Free Shipping Is Not Free: A Data-Driven Model to Design Free-Shipping Threshold Policies. *Available at SSRN 3250971* Preprint, submitted on 2018-10-09.

Chen C, Ngwe D (2018) Shipping Fees and Product Assortment in Online Retail Preprint, submitted on 2018-09-21.

Chen Y (2024) Alibaba's E-Grocer Freshippo to Resume New Membership Subscriptions. URL `https://www.yicaiglobal.com/news/alibabas-e-grocer-freshippo-to-resume-new-membership-subscriptions`.

Choudhary V, Shunko M, Netessine S (2021) Does immediate feedback make you not try as hard? a study on automotive telematics. *Manufacturing & Service Operations Management* 23(4):835–853.

Cooley R, Mobasher B, Srivastava J (1999) Data preparation for mining world wide web browsing patterns. *Knowledge and information systems* 1(1):5–32.

Cui R, Li M, Li Q (2020) Value of high-quality logistics: Evidence from a clash between sf express and alibaba. *Management Science* 66(9):3879–3902.

Cui R, Lu Z, Sun T, Golden JM (2024) Sooner or later? promising delivery speed in online retail. *Manufacturing & Service Operations Management* 26(1):233–251.

Fisher ML, Gallino S, Xu JJ (2019) The value of rapid delivery in omnichannel retailing. *Journal of Marketing Research* 56(5):732–748.

Gümüş M, Li S, Oh W, Ray S (2013) Shipping fees or shipping free? a tale of two price partitioning strategies in online retailing. *Production and operations management* 22(4):758–776.

Guo F, Liu Y (2023) The effectiveness of membership-based free shipping: An empirical investigation of consumers' purchase behaviors and revenue contribution. *Journal of Marketing* 87(6):869–888.

He C, Ozturk OC, Gu C, Chintagunta PK (2023) Consumer tax credits for evs: Some quasi-experimental evidence on consumer demand, product substitution, and carbon emissions. *Management Science* 69(12):7759–7783.

Hemmati S, Elmaghraby WJ, Kabra A, Jain N (2021) Contingent Free Shipping: Drivers of Bubble Purchases. *SSRN Electronic Journal* ISSN 1556-5068, URL `http://dx.doi.org/10.2139/ssrn.3794442`, preprint, submitted on 2021-04-12.

Hu C, Long X, Sun J, Zhang D (2024) Green e-commerce: Environmental impact of fast delivery. *Available at SSRN 4762848* Preprint, submitted on 2024-05-28.

Iacus S, King G, Porro G (2009) Cem: Software for coarsened exact matching. *Journal of statistical Software* 30:1–27.

Jin F, Gao F, Li J (2023) Impact of free shipping threshold on pc and mobile channels: Evidence from an online retailer. *Available at SSRN 3426616* Preprint, submitted on 2023-07-20.

Kabra A, Elmaghraby WJ, Jain N (2024) The role of ease-of-return experience on contingent free shipping policies. *Available at SSRN 4855191* Preprint, submitted on 2024-06-06.

Kahneman D, Tversky A (1979) Prospect theory: An analysis of decision under risk. *Econometrica* 47(2):363–391.

Kanuri VK, Crecelius AT, Kumar S (2025) Disentangling the customer-level, cross-channel effects of large-order-advantaged online shipping policies. *MIS Quarterly* 49(1).

Kivetz R, Urminsky O, Zheng Y (2006) The goal-gradient hypothesis resurrected: Purchase acceleration, illusionary goal progress, and customer retention. *Journal of marketing research* 43(1):39–58.

Koukova NT, Srivastava J, Steul-Fischer M (2012) The effect of shipping fee structure on consumers' online evaluations and choice. *Journal of the Academy of Marketing Science* 40(6):759–770.

Kumar A, Mehra A, Kumar S (2019) Why do stores drive online sales? evidence of underlying mechanisms from a multichannel retailer. *Information Systems Research* 30(1):319–338.

Leng M, Becerril-Arreola R (2010) Joint pricing and contingent free-shipping decisions in b2c transactions. *Production and Operations Management* 19(4):390–405.

Lewis M (2006) The effect of shipping fees on customer acquisition, customer retention, and purchase quantities. *Journal of Retailing* 82(1):13–23.

Lewis M, Singh V, Fay S (2006) An empirical study of the impact of nonlinear shipping and handling fees on purchase incidence and expenditure decisions. *Marketing Science* 25(1):51–64.

Li G, Sheng L, Zhan D (2023) Designing shipping policies with top-up options to qualify for free delivery. *Production and Operations Management* 32(9):2704–2722.

Lim SFW, Wang Q, Webster S (2023) Do it right the first time: Vehicle routing with home delivery attempt predictors. *Production and Operations Management* 32(4):1262–1284.

Lu Z, Cui R, Sun T, Wu L (2025) The value of last-mile delivery in online retail. *Available at SSRN 4590356* Preprint, submitted on 2025-06-09.

Mejia J, Mankad S, Gopal A (2021) Service quality using text mining: Measurement and consequences. *Manufacturing & Service Operations Management* 23(6):1354–1372.

Moran CD (2023) Amazon Fresh lowers its free delivery threshold to $100. URL `https://www.grocerydive.com/news/amazon-fresh-lowers-free-grocery-delivery-threshold/695848`, [Accessed 25-07-2025].

Nigrelli C (2025) Some retailers ditch free shipping as tariffs drive up costs. URL `https://san.com/cc/some-retailers-ditch-free-shipping-as-tariffs-drive-up-costs`, [Accessed 2026-02-04].

Robins JM, Hernan MA, Brumback B (2000) Marginal structural models and causal inference in epidemiology.

Robins JM, Rotnitzky A, Zhao LP (1994) Estimation of regression coefficients when some regressors are not always observed. *Journal of the American statistical Association* 89(427):846–866.

Sancia (2025) Emerging trends in e-commerce logistics and fulfillment. URL `https://www.jusdaglobal.com/en/article/emerging-trends-ecommerce-logistics-fulfillment/`.

Seller Commerce (2025a) 51 eCommerce Statistics In 2025 (Global and U.S. Data). URL `https://www.sellerscommerce.com/blog/ecommerce-statistics/`, [Accessed 2025-08-18].

Seller Commerce (2025b) 15 important free shipping statistics in ecommerce. URL `https://www.sellerscommerce.com/blog/ecommerce-statistics/free-shipping-statistics`, [Accessed 2025-11-13].

Shao XF (2017) Free or calculated shipping: Impact of delivery cost on supply chains moving to online retailing. *International Journal of Production Economics* 191:267–277.

Shehu E, Papies D, Neslin SA (2020) Free shipping promotions and product returns. *Journal of Marketing Research* 57(4):640–658.

Shopify (2025) Free shipping guide: 8 strategies to boost sales (2026). URL `https://www.shopify.com/ca/blog/free-shipping-and-conversion`, [Accessed 2025-12-27].

Statista (2025) Revenue share of the e-commerce market worldwide from 2017 to 2030, by sales channel. URL `https://www.statista.com/statistics/534123/e-commerce-share-of-retail-sales-worldwide`.

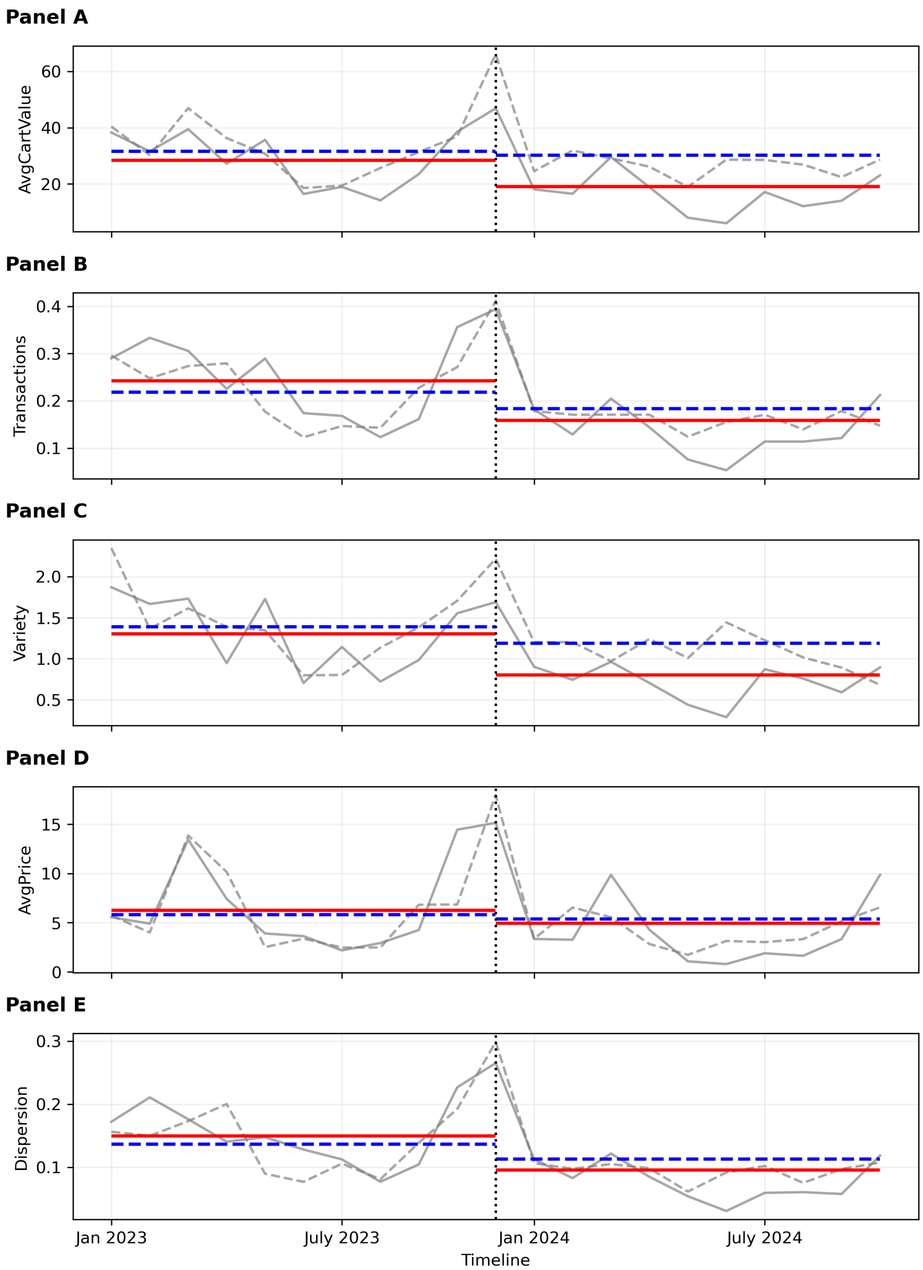
Tereyağoğlu N, Fader PS, Veeraraghavan S (2018) Multiattribute loss aversion and reference dependence: Evidence from the performing arts industry. *Management Science* 64(1):421–436.

USPS Delivery (2025) Order minimums make free shipping beneficial. URL `https://www.uspsdelivers.com/implementing-free-shipping-through-minimum-order-value`, [Accessed 2025-12-27].

Electronic Companion

EC.1. Model-free Evidence

Figure EC.1 Model-free Evidence: Time Series of Outcome Variables During the Observation Period



Notes: The grey dashed line represents the time series of the average outcome variable for the control group, and the grey solid line represents the same for the treatment group. The vertical dotted line marks the beginning of the post-intervention period. The blue dashed horizontal lines indicate the pre- and post-intervention averages for the control group, and the red solid horizontal lines indicate those for the treatment group.

EC.2. Heterogeneous Treatment Effects

EC.2.1. Neighborhood Grocery Alternatives

We measure the intensity of offline grocery alternatives using the number of grocery stores within each consumer's neighborhood, obtained via the Google API. Specifically, we first use the Google Geocoding API to retrieve the latitude and longitude associated with each postal code. Next, we employ the Google Places API to count the number of offline grocery stores located within the neighborhood of each postal code. The neighborhood radius is defined as 1 km for urban areas and 3 km for rural areas, and both are accessible within 10 minutes. We then classify consumers into two groups: $HighAlternative = 1$ if the number of nearby stores exceeds the median value, and $HighAlternative = 0$ otherwise.

We estimate the heterogeneous treatment effects by re-estimating Specification (1) separately for the two groups. The results in Table EC.1 show that the CFS threshold increase significantly reduced total spending and average cart value among consumers with high local alternatives. In contrast, the effects for those with fewer alternatives are statistically insignificant. This pattern suggests that greater offline grocery accessibility amplifies the negative impact of stricter shipping thresholds on online spending, consistent with consumers switching to offline alternatives.

Table EC.1 Effects of CFS Threshold Increase on Consumer Shopping Pattern.

	HighAlternative=0			*HighAlternative*=1		
	Spending	*AvgCartValue*	*Transactions*	*Spending*	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.071 (0.155)	-0.076 (0.152)	-0.003 (0.024)	-0.363*** (0.132)	-0.346*** (0.129)	-0.060*** (0.020)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2430	2430	2430	2718	2718	2718
Adjusted R^2	0.03	0.03	0.03	0.03	0.03	0.03

*Notes: Standard errors are clustered at the consumer-level. All variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

EC.2.2. Neighborhood Affluence

To operationalize neighborhood affluence, we use median after-tax household income at the census tract level as a proxy for local purchasing power. Each user is assigned the median income of their corresponding tract based on postal code linkage. We then define *HighAffluence*= 1 for consumers whose neighborhood median income exceeds the sample median, and *HighAffluence*= 0 otherwise. The results in Table EC.2 indicate that the treatment effect is strong among consumers in low-income neighborhoods. For these consumers, both total spending and average cart value decline significantly after the policy change, whereas consumers in higher-income areas exhibit negligible changes. This pattern suggests that raising the free-shipping threshold primarily affects consumers with limited budgets.

Table EC.2 Effects of CFS Threshold Increase on Consumer Shopping Pattern.

	HighAffluence= 0			*HighAffluence*= 1		
	Spending	*AvgCartValue*	*Transactions*	*Spending*	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.302** (0.152)	-0.287* (0.149)	-0.047** (0.023)	-0.073 (0.137)	-0.078 (0.134)	-0.007 (0.021)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2193	2193	2193	2955	2955	2955
Adjusted R^2	0.03	0.03	0.03	0.03	0.03	0.03

*Notes: Standard errors are clustered at the consumer-level. All variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

EC.2.3. Neighborhood Household Composition

We proxy household-level demand using the average household size of each census tract. We define *HighHousehold*= 1 if the neighborhood's average household size exceeds the sample median and *HighHousehold*= 0 otherwise. As shown in Table EC.3, consumers in smaller-household neighborhoods exhibit significant declines in all outcome variables following the threshold increase, whereas the effects are statistically insignificant for consumers in larger-household areas.

Table EC.3 Effects of CFS Threshold Increase on Consumer Shopping Pattern.

	HighHousehold= 0			*HighHousehold*= 1		
	Spending	*AvgCartValue*	*Transactions*	*Spending*	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.427** (0.211)	-0.411* (0.206)	-0.060* (0.031)	-0.136 (0.118)	-0.134 (0.115)	-0.020 (0.018)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1258	1258	1258	3890	3890	3890
Adjusted R^2	0.01	0.01	0.02	0.03	0.03	0.03

*Notes: Standard errors are clustered at the consumer-level. All variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

EC.3. Counterfactual Analysis and Threshold Design

EC.3.1. Proof of Proposition 1

Fix $c < 0$, $C > 0$, $k \in [0,1)$, and $\underline{\tau} < \overline{\tau}$. Let the average cart value under threshold τ be

$$y(\tau) = a\left(\frac{\tau}{\underline{\tau}}\right)^{c}, \qquad \tau \in [\underline{\tau}, \overline{\tau}],$$

where $a = y(\underline{\tau})$ is given. The platform's per-order net revenue is

$$\pi(\tau) = y(\tau) + S(y(\tau), \tau) - C,$$

where

$$S(y,\tau)=\begin{cases}C-ky, & y<\tau,\\ 0, & y\geq\tau.\end{cases}$$

We first establish the basic monotonicity of demand. Differentiating $y(\tau)$ yields

$$y'(\tau)=\frac{c}{\tau}y(\tau),$$

which is strictly negative for all $\tau>0$ because $c<0$. Hence $y(\tau)$ is strictly decreasing in τ on $[\underline{\tau},\overline{\tau}]$.

Next, consider the equation $y(\tau)=\tau$. Substituting the expression for $y(\tau)$ gives

$$a\left(\frac{\tau}{\underline{\tau}}\right)^{c}=\tau \quad\Longleftrightarrow\quad \tau^{1-c}=a\,\underline{\tau}^{-c}.$$

Since $1-c>1$, this equation admits a unique positive solution,

$$\hat{\tau}(a)=\left(a\,\underline{\tau}^{-c}\right)^{\frac{1}{1-c}}.$$

Because $y(\tau)$ is strictly decreasing while τ is increasing, we have $y(\tau)>\tau$ if and only if $\tau<\hat{\tau}(a)$ and $y(\tau)<\tau$ if and only if $\tau>\hat{\tau}(a)$.

Using this characterization, the profit function can be written as

$$\pi(\tau)=\begin{cases}y(\tau)-C, & \tau\leq\hat{\tau}(a),\\ (1-k)y(\tau), & \tau>\hat{\tau}(a).\end{cases}$$

On the interval $\tau\leq\hat{\tau}(a)$, i.e., the free shipping region, $\pi(\tau)=y(\tau)-C$ is strictly decreasing in τ. On the interval $\tau>\hat{\tau}(a)$, i.e., the surcharge region, $\pi(\tau)=(1-k)y(\tau)$ is also strictly decreasing in τ because $1-k>0$ and $y(\tau)$ is strictly decreasing. Therefore, within each region the profit function is strictly decreasing in τ.

It follows that any optimal threshold must be attained either at the lower bound $\underline{\tau}$ or at the boundary between the two regions, provided that boundary is feasible. If $\hat{\tau}(a)>\overline{\tau}$, then $y(\tau)\geq\tau$ for all $\tau\in[\underline{\tau},\overline{\tau}]$ and the entire feasible set lies in the free-shipping region. Since $\pi(\tau)=y(\tau)-C$ is strictly decreasing on this interval, the unique maximizer is $\tau^*=\underline{\tau}$.

Suppose instead that $\hat{\tau}(a)\leq\overline{\tau}$. In this case, both the free-shipping region and the surcharge region are feasible. The best feasible threshold in the free-shipping region is $\underline{\tau}$, yielding profit

$$\pi(\underline{\tau})=y(\underline{\tau})-C=a-C.$$

In the surcharge region, because $\pi(\tau)$ is strictly decreasing for $\tau > \hat{\tau}(a)$, the largest attainable profit is achieved in the limit as τ approaches $\hat{\tau}(a)$ from above. By continuity of $y(\tau)$,

$$\lim_{\tau \downarrow \hat{\tau}(a)} \pi(\tau) = (1-k)y(\hat{\tau}(a)) = (1-k)\hat{\tau}(a).$$

Thus, when $\hat{\tau}(a) \le \overline{\tau}$, the platform prefers the kink to the lower bound if and only if

$$(1-k)\hat{\tau}(a) \ge a - C.$$

Define $a^* > 0$ as the unique solution to

$$(1-k)\hat{\tau}(a^*) = a^* - C.$$

Such a solution exists because $\hat{\tau}(a)$ is continuous and strictly increasing in a, grows sublinearly in a, and satisfies $\hat{\tau}(\underline{\tau}) = \underline{\tau}$.

Therefore, if $a \in [\underline{\tau}, a^*]$ and $\hat{\tau}(a) \le \overline{\tau}$, the optimal threshold is $\tau^* = \hat{\tau}(a)$ (interpreted as the right-limit at the kink). In all other cases, the optimal threshold is $\tau^* = \underline{\tau}$. This completes the proof.

EC.3.2. Impact of Shipping Discount

We also vary the discount factor k and investigate the performance of a partially discounted CFS policy. We set $k = 0$ (no discounting), $k = 0.05$ (small discounting), and $k = 0.1$ (large discounting). The results are shown in Figure EC.2. We find that as the discounting factor increases, the support for an optimal threshold above 80 shrinks (Panel A). In addition, the minimum relative improvement increases, indicating that offering consumers a larger discount on the shipping surcharge makes suboptimal threshold choices more costly.

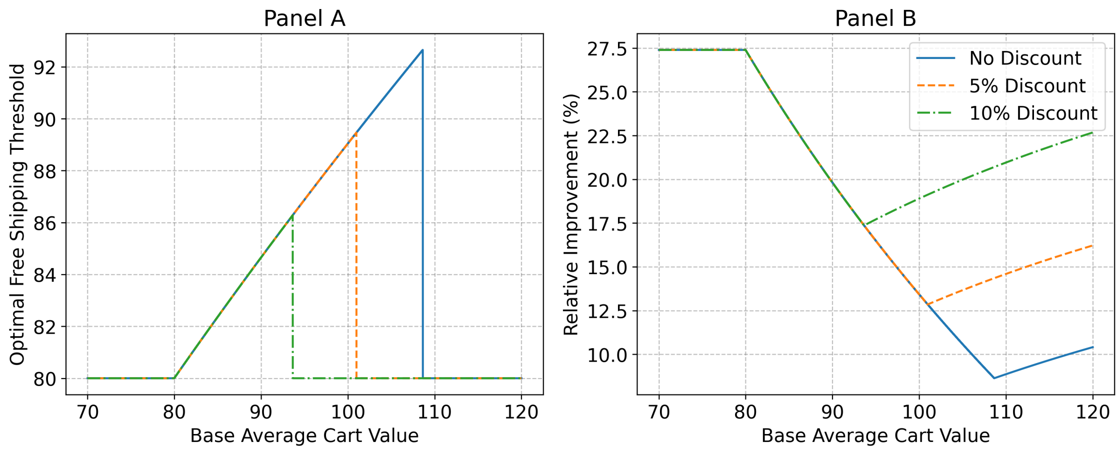
EC.3.3. Stochastic Optimization with Cart Value Heterogeneity

This section extends the deterministic threshold optimization problem by allowing the original cart values and the elasticity to be heterogeneous across consumers and orders. Let $a \equiv y(\underline{\tau})$ denote the cart value at the original free shipping threshold $\underline{\tau}$, and let c denote the (order-level) elasticity of cart value with repsect to the threshold. We assume that (a, c) is a random vector on $\mathbb{R}_+ \times \mathbb{R}$ that jointly follows distribution $\mathcal{F}$, with the cumulative distribution function $F(\cdot, \cdot)$.

For any candidate threshold $\tau \in [\underline{\tau}, \overline{\tau}]$, realized cart value is given by

$$y(\tau; a, c) = a\left(\frac{\tau}{\underline{\tau}}\right)^c,$$

Figure EC.2 **Optimal Threshold Values and Revenue Improvement with Various Base Cart Values and Shipping Discounts**



Panel A plots the optimal free-shipping threshold as a function of the base average cart value. Panel B reports the relative improvement in net consumer revenue (%) when the threshold is set to its optimal value compared with a fixed threshold of \$100, for each base average cart value.

An order qualifies for free shipping if and only if $y(\tau;a,c) \geq \tau$, which is equivalent to

$$a \geq a_0(\tau;c) \equiv \tau^{1-c}\underline{\tau}^c.$$

The platform's per-cart net revenue from an order with baseline cart value a is

$$\pi(\tau;a,c) = \begin{cases} y(\tau;a,c) - C, & a \geq a_0(\tau;c), \\ (1-k)y(\tau;a,c), & a < a_0(\tau;c), \end{cases}$$

where C is the per-cart fulfillment cost and $k \in [0,1)$ governs how the shipping surcharge varies with cart value. The platform chooses τ to maximize expected per-cart profit,

$$\Pi(\tau) \equiv \mathbb{E}_{\mathcal{F}}[\pi(\tau;a,c)], \qquad \tau^* \in \arg\max_{\underline{\tau} \leq \tau \leq \overline{\tau}} \Pi(\tau)$$

Assumption EC.1. *(1) The elasticity c is a constant and negative. (2) The original cart value follows a uniform distribution, $a \sim Unif[L,U]$, where $0 < L < U$.*

Note that the first assumption is supported by the estimation results from a random coefficient model in Section EC.3.3.1. The second assumption simplifies the analysis. Despite its simplicity, the uniform case highlights that the optimal free-shipping threshold depends not only on average cart value but also on its dispersion.

Let $g(\tau) \equiv (\tau/\underline{\tau})^c$. Under the above assumption, we have

$$y(\tau;a,c) = g(\tau)a, \qquad F(t) = \frac{t-L}{U-L}, \qquad \mathbb{E}[a\,\mathbf{1}\{a \geq t\}] = \frac{U^2 - t^2}{2(U-L)}.$$

Substituting $t = a_0(\tau)$ yields

$$\Pi(\tau) = g(\tau)\left[(1-k)\frac{L+U}{2} + k\frac{U^2 - a_0(\tau)^2}{2(U-L)}\right] - C\left(1 - \frac{a_0(\tau) - L}{U-L}\right),$$

for $a_0(\tau) \in (L,U)$. Let $\mu \equiv \mathbb{E}[a] = (L+U)/2$. When $a_0(\tau) \le L$, all orders qualify for free shipping and $\Pi(\tau) = g(\tau)\mu - C$; when $a_0(\tau) \ge U$, no orders qualify and $\Pi(\tau) = (1-k)g(\tau)\mu$.

Because $g(\tau)$ is strictly decreasing in τ, expected profit is decreasing in τ outside the interior region $a_0(\tau) \in (L,U)$. Hence, an interior optimum (if it exists) must lie in the mixed regime where some orders qualify for free shipping and others incur a surcharge. In this region, the optimal threshold trades off the marginal loss from reduced demand against the marginal gain from shifting probability mass from free shipping to surcharge, with the density of orders around $a_0(\tau)$ playing a central role.

PROPOSITION EC.1 **(Stochastic Optimal Free-Shipping Thresholds)**. *Let $D \equiv U - L$ and $\mu \equiv \mathbb{E}[a] = (L+U)/2$ under Assumption EC.1. Define the cutoff*

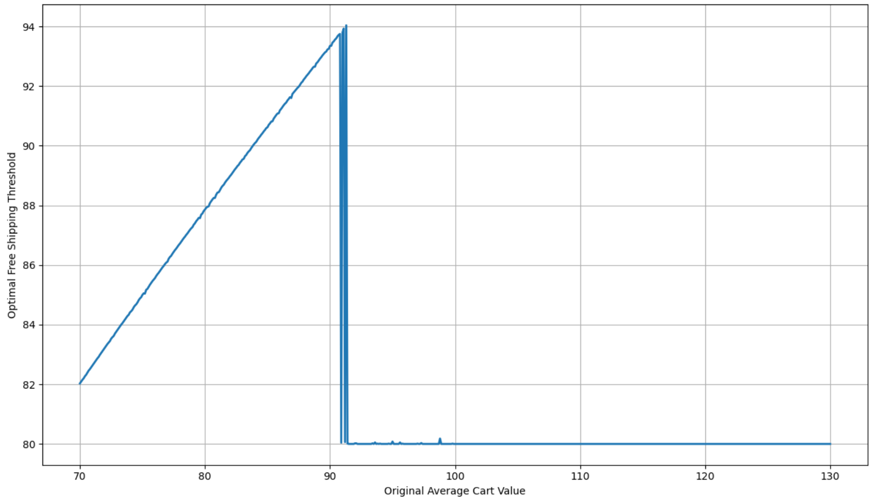
$$\mu^* \;\equiv\; \frac{1}{1-k}\left[\frac{1-c}{-c}\cdot\frac{\underline{\tau}}{D}\big(C - k\underline{\tau}\big) \;-\; k\cdot\frac{U^2-\underline{\tau}^2}{2D}\right].$$

If $\mu < \mu^*$, then the platform optimally increases *the threshold above $\underline{\tau}$.*

The proof can be seen in EC.3.3.2. We highlight that Proposition EC.1 naturally extends our findings in the main body (Proposition 1) to a stochastic setting with mild assumptions. In the stochastic model, the platform chooses a single threshold τ for a population of heterogeneous orders. Increasing τ raises $a_0(\tau)$, thereby shifting some orders from the free shipping regime into the surcharge regime. Unlike the deterministic case, the marginal benefits depends on the amount of probability mass near the cufoff $a_0(\tau)$, i.e., on the distribution of original cart value $\mathcal{F}$, not just its mean. Under Assumption EC.1, the strength of this shift is reciprocal to dispersion D. This is why the cutoff μ^* in Proposition EC.1 depends on both the mean μ and dispersion D. In particular, holding μ fixed, greater dispersion lowers the density around the cutoff and weakens the incentive to raise the threshold τ.

Figure EC.3 illustrates these results numerically. We assume that pre-policy cart values follow a uniform distribution, $a \sim \mathrm{Unif}[0.8\mu, 1.2\mu]$, where $\mu = \mathbb{E}[a]$ varies from 70 to 130. For each value of μ, we draw 10,000 realizations of a and compute the optimal free-shipping threshold by maximizing empirical per-cart profit. Consistent with Proposition EC.1, the optimal policy exhibits a cutoff structure: increasing the threshold above the baseline is optimal only for μ below a critical value, while for larger μ the optimal threshold remains at the lower bound and should not be increased further.

Figure EC.3 **Optimal Threshold Values and Revenue Improvement with Various Base Cart Values**



EC.3.3.1. A random-slope DiD model. We assume the treatment effect is random at the consumer level, and estimate the coefficients with the following DiD specification:

$$\log(AvgCartValue + 1) = \alpha_0 + (\alpha_1 + \alpha_{1,i}) \cdot Treated_i \times Post_t + \mu_i + \delta_t + \varepsilon_{it}$$

where α_1 captures the common treatment effect, and $\alpha_{1,i} \sim \mathcal{N}(0, \sigma^2)$ captures the random heterogeneous treatment effects across consumers. Table EC.4 reports the results. In particular, $\alpha_1 = -0.166$ and $\sigma^2 \simeq 0$, which support the the first part in Assumption EC.1.

Table EC.4 **Random Effects of CFS Threshold Increase on Average Cart Value**

	(1) *AvgCartValue*
$Treated \times Post$	-0.166***
	0.064
σ^2	2.8e-14
Consumer Fixed Effects	Yes
Yearmonth Fixed Effects	Yes
Observations	5148
Log-likelihood	-10464.43

*Notes: Standard errors are clustered at the consumer-level. * p<0.10, ** p<0.05, *** p<0.01.*

EC.3.3.2. Proof of Proposition EC.1. Fix $\tau \in [\underline{\tau}, \overline{\tau}]$. Let

$$g(\tau) \equiv \left(\frac{\tau}{\underline{\tau}}\right)^c, \qquad a_0(\tau) \equiv \tau^{1-c}\underline{\tau}^c,$$

so that $y(\tau; a) = g(\tau)a$ and $y(\tau; a) \geq \tau$ if and only if $a \geq a_0(\tau)$. Note that $g(\tau) > 0$ and, since $c < 0$, $g'(\tau) = \frac{c}{\tau} g(\tau) < 0$, while $a_0'(\tau) = \frac{1-c}{\tau} a_0(\tau) > 0$. Moreover,

$$g(\tau)\, a_0(\tau) = \tau. \tag{EC.1}$$

The platform maximizes expected per-order profit $\Pi(\tau) \equiv \mathbb{E}[\pi(\tau;a)]$, where

$$\pi(\tau;a) = \begin{cases} y(\tau;a) - C, & a \geq a_0(\tau), \\ (1-k)y(\tau;a), & a < a_0(\tau). \end{cases}$$

Using $y(\tau;a) = g(\tau)a$ and $a \sim \text{Unif}[L,U]$, we obtain

$$\begin{aligned} \Pi(\tau) &= \mathbb{E}\Big[(g(\tau)a - C)\mathbf{1}\{a \geq a_0(\tau)\} + (1-k)g(\tau)a\mathbf{1}\{a < a_0(\tau)\}\Big] \\ &= g(\tau)\Big((1-k)\mathbb{E}[a] + k\,\mathbb{E}[a\mathbf{1}\{a \geq a_0(\tau)\}]\Big) - C\,\mathbb{P}(a \geq a_0(\tau)). \end{aligned}$$

In the mixed region $a_0(\tau) \in (L,U)$, the uniform distribution yields

$$\mathbb{P}(a \geq t) = \frac{U-t}{D}, \qquad \mathbb{E}[a\mathbf{1}\{a \geq t\}] = \frac{U^2 - t^2}{2D}.$$

Substituting $t = a_0(\tau)$ gives, for $a_0(\tau) \in (L,U)$,

$$\Pi(\tau) = g(\tau)\left[(1-k)\mu + k\frac{U^2 - a_0(\tau)^2}{2D}\right] - C\frac{U - a_0(\tau)}{D}. \tag{EC.2}$$

We next compute the derivative at the baseline threshold $\underline{\tau}$. Since $\underline{\tau}$ lies in the mixed region, (EC.2) is differentiable at $\underline{\tau}$. Differentiating (EC.2) and using $g'(\tau) = \frac{c}{\tau}g(\tau)$, $t'(\tau) = \frac{1-c}{\tau}a_0(\tau)$, and the identity (EC.1), yields

$$\Pi'(\tau) = \frac{1}{\tau}\left\{c\,g(\tau)\left[(1-k)\mu + k\frac{U^2 - a_0(\tau)^2}{2D}\right] + (1-c)\frac{a_0(\tau)}{D}\big(C - k\tau\big)\right\}, \qquad a_0(\tau) \in (L,U).$$

Evaluating at $\tau = \underline{\tau}$ gives $g(\underline{\tau}) = 1$ and $t(\underline{\tau}) = \underline{\tau}$, hence

$$\Pi'(\underline{\tau}) = \frac{1}{\underline{\tau}}\left\{c\left[(1-k)\mu + k\frac{U^2 - \underline{\tau}^2}{2D}\right] + (1-c)\frac{\underline{\tau}}{D}\big(C - k\underline{\tau}\big)\right\}. \tag{EC.3}$$

Because $\underline{\tau} > 0$, the sign of $\Pi'(\underline{\tau})$ equals the sign of the bracketed expression. Rearranging (EC.3) shows that $\Pi'(\underline{\tau}) \leq 0$ if and only if $\mu \geq \mu^*$, where

$$\mu^* = \frac{1}{1-k}\left[\frac{1-c}{-c}\cdot\frac{\underline{\tau}}{D}\big(C - k\underline{\tau}\big) - k\cdot\frac{U^2 - \underline{\tau}^2}{2D}\right].$$

Finally, since $\Pi(\tau)$ is continuous on the compact set $[\underline{\tau}, \overline{\tau}]$, a global maximizer τ^* exists. If $\Pi'(\underline{\tau}) > 0$, then $\Pi(\tau) > \Pi(\underline{\tau})$ for τ in a neighborhood above $\underline{\tau}$, so $\underline{\tau}$ cannot be the global maximizer and $\tau^* > \underline{\tau}$. Combining with the equivalence $\Pi'(\underline{\tau}) > 0 \Longleftrightarrow \mu < \mu^*$ yields

$$\mu < \mu^* \implies \tau^* > \underline{\tau}.$$

This completes the proof.

EC.3.4. A Fully Interacted DiD Model

The fully interacted DiD model specification with both the reference point effect and search effort is as follows:

$$\begin{aligned}\log(\text{AvgCartValue}_{it}+1) = \beta_0 + \mu_i + \delta_t + \beta_1(\text{Treated}\times\text{Post}) \\ + \beta_2(\text{Treated}\times\text{Post}\times\text{Recent}) + \beta_3(\text{Recent}\times\text{Post}) \\ + \beta_4(\text{Treated}\times\text{Post}\times\text{HighEffort}) + \beta_5(\text{HighEffort}\times\text{Post})\end{aligned}$$

where β_1 captures the effect on baseline consumers who are neither recently exposed to the old threshold nor subject to high search effort, and β_2 and β_4 capture the additional impacts due to the reference point effect and search effort, respectively. The estimated results are shown in Table EC.5.

Table EC.5 Heterogeneous Effects of CFS Threshold Increase on Consumer Search Effort.

	AvgCartValue
Treated × Post	0.196*
	(0.116)
Treated × Post × Recent	-0.503***
	(0.180)
Treated × Post × HighEffort	-0.488**
	(0.208)
Consumer Fixed Effects	Yes
Yearmonth Fixed Effects	Yes
Observations	5148
Adjusted R^2	0.04

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01.*

The elasticity parameter can be obtained as follows.

$$\hat{c}_{\text{baseline}} = \frac{\hat{\beta}_1}{\log(100/80)} = 0.88$$

$$\hat{c}_{\text{reference}} = \frac{\hat{\beta}_1+\hat{\beta}_2}{\log(100/80)} = -1.39$$

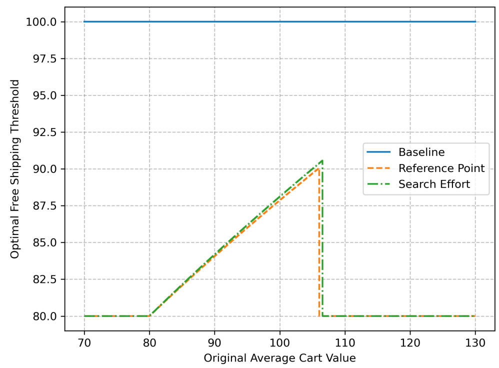
$$\hat{c}_{\text{search}} = \frac{\hat{\beta}_1+\hat{\beta}_4}{\log(100/80)} = -1.31$$

The optimal thresholds for various consumer segments are shown in Figure EC.4.

EC.3.5. DiD Analysis with Continuous Interpurchase Time

We denote IntraTime_i as a continuous variable, indicating the time difference between consumer i's last pre-treatment purchase and the threshold change. We run the following DiD specification. The estimated results are shown in Table EC.6.

Figure EC.4 Optimal Threshold Values with Various Consumer Segments



$$\log(\text{AvgCartValue}_{it}+1) = \beta_0 + \mu_i + \delta_t + \beta_1(\text{Treated} \times \text{Post})$$
$$+ \beta_2 \text{Treated}_i \times \text{Post}_t \times \log(\text{IntraTime}_i) + \beta_3 \text{Post}_t \times \log(\text{IntraTime}_i)$$

Table EC.6 c Effects of CFS Threshold Increase on Consumer Search Effort.

	AvgCartValue
Treated × Post	-0.674***
	(0.158)
Treated × Post × IntraTime	0.380***
	(0.080)
Consumer Fixed Effects	Yes
Yearmonth Fixed Effects	Yes
Observations	5148
Adjusted R^2	0.03

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01.*

EC.4. Estimation Results for Robustness Checks

EC.4.1. Parallel Trends Assumption

Table EC.7 Regression Results for the Parallel Trends Assumption Tests.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times 202302$	0.359 (0.417)	0.354 (0.415)	0.060 (0.063)	0.160 (0.160)	0.143 (0.260)	0.047 (0.063)
$Treated \times 202303$	0.056 (0.426)	0.045 (0.423)	0.018 (0.062)	0.090 (0.164)	-0.079 (0.282)	-0.013 (0.062)
$Treated \times 202304$	-0.293 (0.411)	-0.295 (0.406)	-0.041 (0.062)	-0.066 (0.158)	-0.296 (0.270)	-0.079 (0.060)
$Treated \times 202305$	0.478 (0.412)	0.471 (0.407)	0.077 (0.061)	0.216 (0.173)	0.220 (0.241)	0.037 (0.054)
$Treated \times 202306$	0.188 (0.396)	0.183 (0.392)	0.034 (0.060)	0.087 (0.161)	0.049 (0.241)	0.032 (0.059)
$Treated \times 202307$	0.100 (0.373)	0.106 (0.372)	0.014 (0.055)	0.122 (0.149)	-0.014 (0.227)	-0.019 (0.053)
$Treated \times 202308$	-0.054 (0.388)	-0.041 (0.387)	-0.005 (0.056)	-0.009 (0.162)	-0.036 (0.232)	-0.030 (0.053)
$Treated \times 202309$	-0.238 (0.420)	-0.239 (0.418)	-0.037 (0.061)	-0.057 (0.170)	-0.221 (0.259)	-0.057 (0.060)
$Treated \times 202310$	0.242 (0.446)	0.227 (0.442)	0.044 (0.068)	0.035 (0.175)	0.177 (0.289)	0.006 (0.066)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2277	2277	2277	2277	2277	2277
Adjusted R^2	0.02	0.02	0.02	0.02	0.02	0.02

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except Dispersion, are log-transformed. We use only the pre-intervention observations.*

EC.4.2. Placebo Tests

Table EC.8 Regression Results for the Placebo Tests.

	(1) *Expenditure*	(2) *AvgCartValue*	(3) *Transactions*	(4) *SKUs*	(5) *AvgPrice*	(6) *PopShare*
Panel A: Placebo test with fake CFS threshold increase.						
$Treated \times PlaceboPost$	-0.074 (0.204)	-0.069 (0.201)	-0.013 (0.031)	-0.046 (0.082)	-0.006 (0.125)	-0.011 (0.030)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2277	2277	2277	2277	2277	2277
Adjusted R^2	0.03	0.03	0.04	0.08	0.02	0.04
Panel B: Placebo test with fake treatment group						
$PlaceboTreated \times Post$	-0.142 (0.163)	-0.136 (0.161)	-0.021 (0.024)	-0.049 (0.065)	-0.082 (0.100)	-0.001 (0.023)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2549	2549	2549	2549	2549	2549
Adjusted R^2	0.03	0.03	0.04	0.05	0.03	0.04

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except Dispersion, are log-transformed. We use only the observations of the control group.*

EC.4.3. CEM

We first show that, despite the non-random assignment, consumers' browsing and purchasing behaviors are largely statistically indistinguishable between the treatment and control groups prior to the intervention. We consider six outcome variables described in Section 3.2.3, along with four

Table EC.9 Covariate Balance Between Treatment and Control Groups Before Matching.

	Treatment Group		Control Group			
Variable	Mean	Std	Mean	Std	t-stats	p-value
Spending	5.306	0.773	5.373	0.794	-0.689	0.491
AvgCartValue	4.760	0.474	4.882	0.536	-1.942	0.053*
Transactions	1.044	0.396	1.002	0.343	0.922	0.357
Variety	1.980	0.884	2.069	0.911	-0.801	0.424
AvgPrice	3.465	0.826	3.465	0.692	-0.007	0.995
PopShare	0.504	0.176	0.496	0.181	0.348	0.728
Sessions	1.279	0.572	1.355	0.706	-0.959	0.338
AvgViews	0.897	0.715	0.933	0.740	-0.398	0.691
AvgSessionDuration	6.620	1.447	6.642	1.510	-0.122	0.903
Conversion	0.572	0.159	0.553	0.166	0.928	0.354

*Notes: * $p<0.10$, ** $p<0.05$, *** $p<0.01$.*

additional measures of browsing behavior. Specifically, *Session* denotes the number of sessions a consumer has in each month; *AvgViews* is the average number of distinct products viewed per session; *AvgSessionDuration* measures the average session length in seconds; and *Conversion* is the proportion of viewed products that result in a purchase. As reported in Table EC.9, none of these variables exhibits statistically significant differences across groups, with the exception of average cart value, which is marginally different.

Nevertheless, we further implement CEM using *Spending*, *AvgCartValue*, *Variety*, and *Conversion* as matching covariates. These variables are selected based on their theoretical relevance for post-treatment purchasing behavior and their relatively low redundancy in the data's correlation structure. Together, they capture the primary dimensions of consumers' purchasing intensity, basket composition, and purchase propensity, while avoiding excessive overlap with highly collinear outcome variables and browsing measures.

We coarsen each covariate into a small number of substantively meaningful bins to maximize common support. Specifically, *Spending* is grouped into three categories (low, medium, and high spenders); *AvgCartValue* into two categories (low versus high); *Variety* into two categories (low versus high); and *Conversion* into two categories (low versus high). This coarse partitioning preserves a large share of treated and control observations in the matched sample while ensuring balance along key pre-treatment behavioral dimensions.

The resulting matched sample exhibited strong overlap and excellent covariate balance. We retain almost 80% of the samples and dropped 19 consumers from the treatment group and 25 from the control group. We report the standardized mean differences (SMD) between groups on all matched covariates to demonstrate the quality of the matching (Table EC.10). The SMD values are reduced across all covariates, well within conventional thresholds for acceptable balance.

Table EC.10 The Quality for the CEM Matching

	SMD	
Covariate	**All Samples**	**Matched Samples**
Spending	-0.130	-0.064
AvgCartValue	0.019	0.031
Variety	-0.121	0.060
Conversion	0.019	-0.015

Table EC.11 reports the treatment effect estimates based on the weighted DiD analysis. The results are consistent with our main findings, providing additional support for the robustness of our conclusions.

Table EC.11 DiD Analysis on Matched Samples.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times Post$	-0.326***	-0.376**	-0.062**	-0.138**	-0.230**	-0.057**
	(0.092)	(0.149)	(0.024)	(0.061)	(0.093)	(0.023)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4280	4280	4280	4280	4280	4280
Adjusted R^2	0.01	0.01	0.01	0.01	0.02	0.02

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except Dispersion, are log-transformed. 19 consumers in the treatment group and 25 in the control group are dropped due to matching, resulting in a smaller sample size.*

EC.4.4. IPW Estimator

For the IPW estimator, we use sample of consumers who placed at least one order before the intervention, which includes consumers who has not placed any order after the intervention ($Survive_i = 0$) and those who have placed at least one order ($Survive = 1$). The expanded data contains 1124 consumers with 2319 completed orders. We use a logit regression to estimate the probability that a consumer will place orders after the intervention based on pre-intervention purchasing and browsing behavior. Specifically,

$$\begin{aligned} \mathbb{P}(Survive_i = 1) = \alpha_0 + \alpha_1 \cdot AvgCartValue_i + \alpha_2 \cdot Transactions_i + \alpha_3 \cdot AvgPrice_i \\ + \alpha_4 \cdot Variety_i + \alpha_5 \cdot PopShare_i + \alpha_6 \cdot Sessions_i \\ + \alpha_7 \cdot AvgViews_i + \alpha_8 \cdot AvgSessionDuration_i + \alpha_9 \cdot Conversion_i + \varepsilon_i \end{aligned} \tag{EC.4}$$

We use the proportion of consumers who have post-intervention orders to approximate $\mathbb{P}(Survive = 1)$, and then calculate the IPW weight for each consumer using equation (5). We further report the standardized mean differences (SMD) between groups on all covariates before and after weighting to demonstrate quality of the probabilities and the balance they deliver once turned into weights.

Table EC.12 The Quality for the IPW weights

Covariate	SMD	
	Unweighted	Weighted
AvgCartValue	0.281	0.012
Transactions	0.215	0.004
AvgPrice	-0.075	-0.031
Variety	0.317	0.006
PopShare	0.096	0.001
Sessions	0.138	-0.002
AvgViews	0.206	0.025
AvgSessionDuration	0.253	0.020
Conversion	-0.187	0.005

As shown in Table EC.12, the absolute SMD before weighting is approximately 0.32, while it significantly reduced after weighting, implying the balance of the IPW weights.

Using these stabilized weights, we re-estimate our DiD model in Specification (1) to recover treatment effects that are representative of the broader population, including consumers who have not placed any orders during the post-intervention period. Table EC.13 summarizes the estimation results.

Table EC.13 Regression Results Using Inverse Probability Weighting.

	(1) *Expenditure*	(2) *AvgCartValue*	(3) *Transactions*	(4) *SKUs*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times Post$	-0.274**	-0.265**	-0.041**	-0.101**	-0.137*	-0.034*
	(0.120)	(0.118)	(0.019)	(0.046)	(0.078)	(0.019)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5148	5148	5148	5148	5148	5148
Adjusted R^2	0.02	0.02	0.01	0.03	0.01	0.03

*Notes: Standard errors are clustered at the consumer level. All outcome variables, except Dispersion, are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

EC.4.5. Aggregation-level Analysis

We consider the following outcome variables, closely aligned with those defined in Section 3.2.3: (1) $Sales_{jt}$ denotes the total value of completed orders from all consumers in borough j during year-month t; (2) $AvgCartValue_{jt}$ captures the average order size; (3) $Transactions_{jt}$ represents the total number of completed orders; (4) $Variety_{jt}$ measures the total number of distinct products purchased, indicating borough-level purchase variety; (5) $AvgPrice_{jt}$ reflects the average unit price of purchased items; and (6) $PopShare_{jt}$ captures the share of popular products among all items purchased, indicating shifts in consumer preference between popular and long-tail products. The aggregation provides a balanced panel dataset of 6040 observations. We show the descriptive statistics in Table EC.14.

Table EC.14 Borough-aggregation-level Variable statistics

Outcome Variable	Observations	mean	std	min	max
Sales	6040	71.26	316.92	0	7265.63
AvgCartValue	6040	30.36	72.18	0	721.57
Transactions	6040	0.53	2.46	0	50
Variety	6040	2.51	10.76	0	191
AvgPrice	6040	6.31	20.99	0	400
PopShare	6040	0.14	0.30	0	1

We use the following model specifications.

$$y_{it} = \alpha_0 + \alpha_1 \cdot Treated_j \times Post_t + \mu_j + \delta_t + \varepsilon_{jt} \tag{EC.5}$$

where $Treated_j = 1$ if borough j is subject to the CFS policy and 0 otherwise. μ_j controls for borough-specific unobserved heterogeneity.

Table EC.15 Regression Results Using Aggregated Borough-Month-Level Panel Data.

	(1) *Sales*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times Post$	-0.390** (0.184)	-0.396** (0.166)	-0.049 (0.039)	-0.209** (0.085)	-0.190* (0.111)	-0.058** (0.028)
Borough Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6040	6040	6040	6040	6040	6040
Adjusted R^2	0.38	0.31	0.59	0.47	0.28	0.23

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except PopShare, are log-transformed.*

Table EC.15 reports the regression results. The estimated coefficients on the interaction term $Treated \times Post$ are significantly negative for total sales and average cart value, suggesting that the policy change led to a meaningful reduction in overall sales and average cart value at the borough level. The effect on transaction volume is negative but not statistically significant, which reflects the finding that transactions contribute marginally in our main results. In addition, we find significant declines in purchase variety, average product price, and the share of popular products purchased. These findings closely mirror our consumer-level results and further reinforce the conclusion that the increase in the CFS threshold discouraged consumer engagement, both in terms of how much they bought and what they chose to buy.